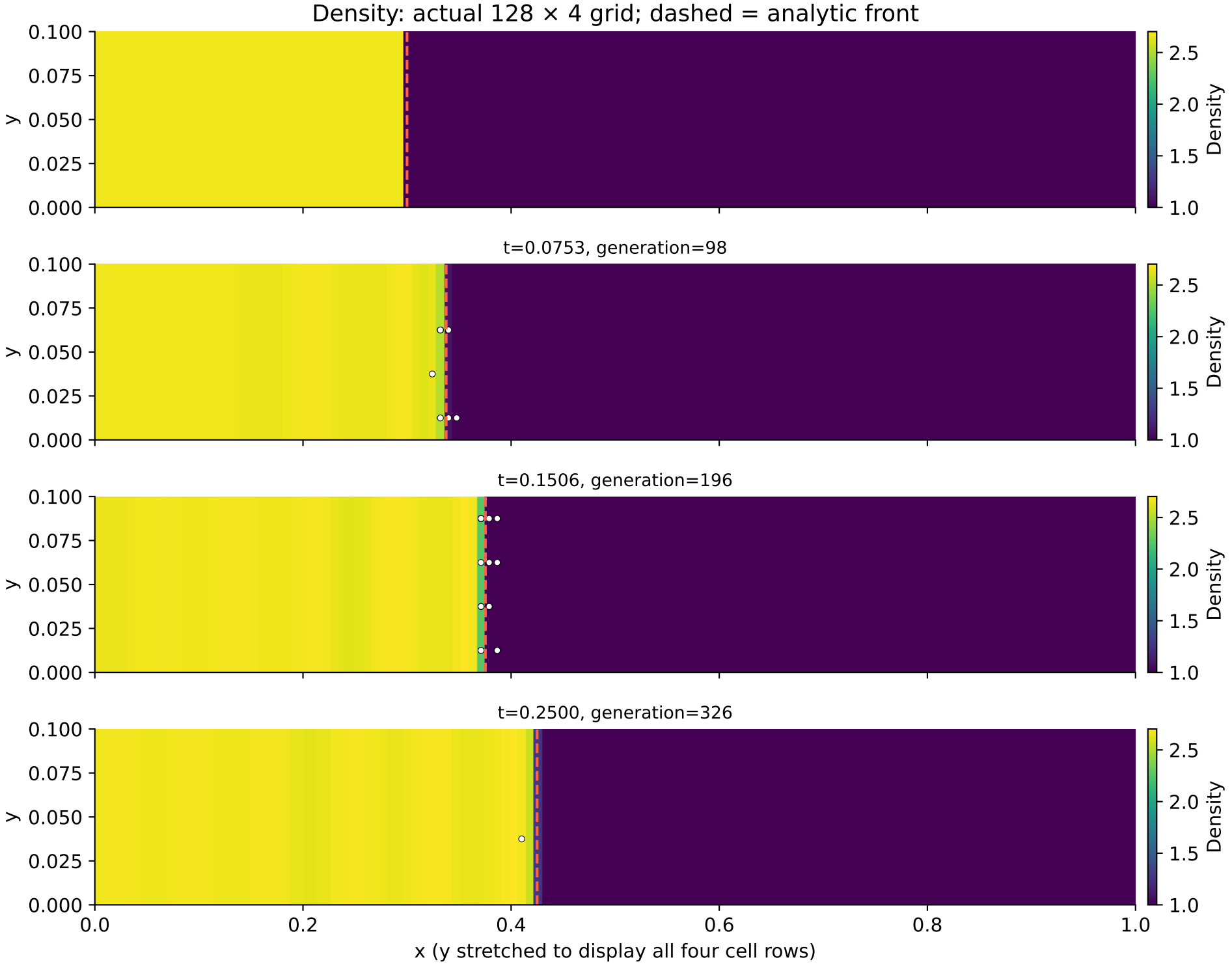
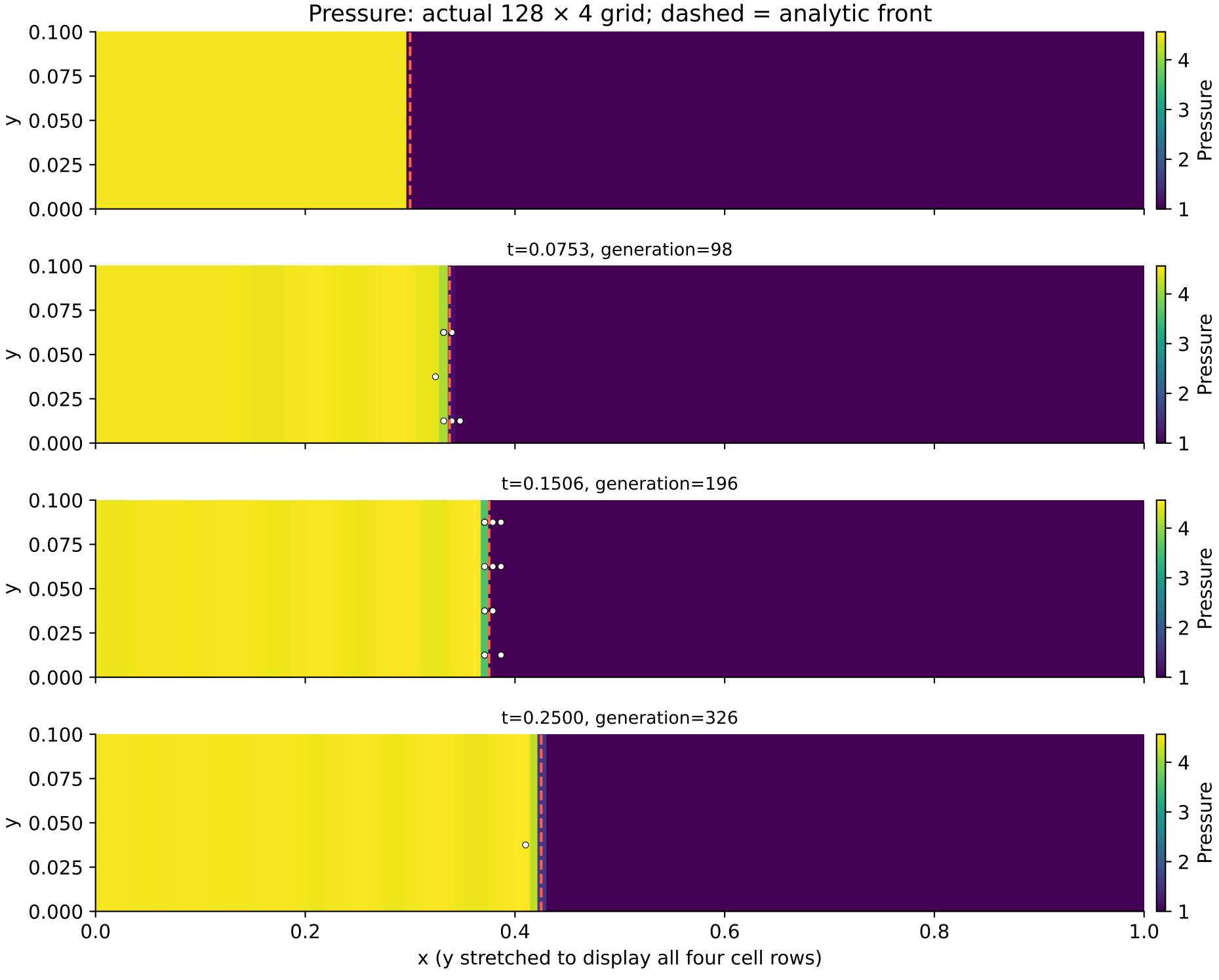
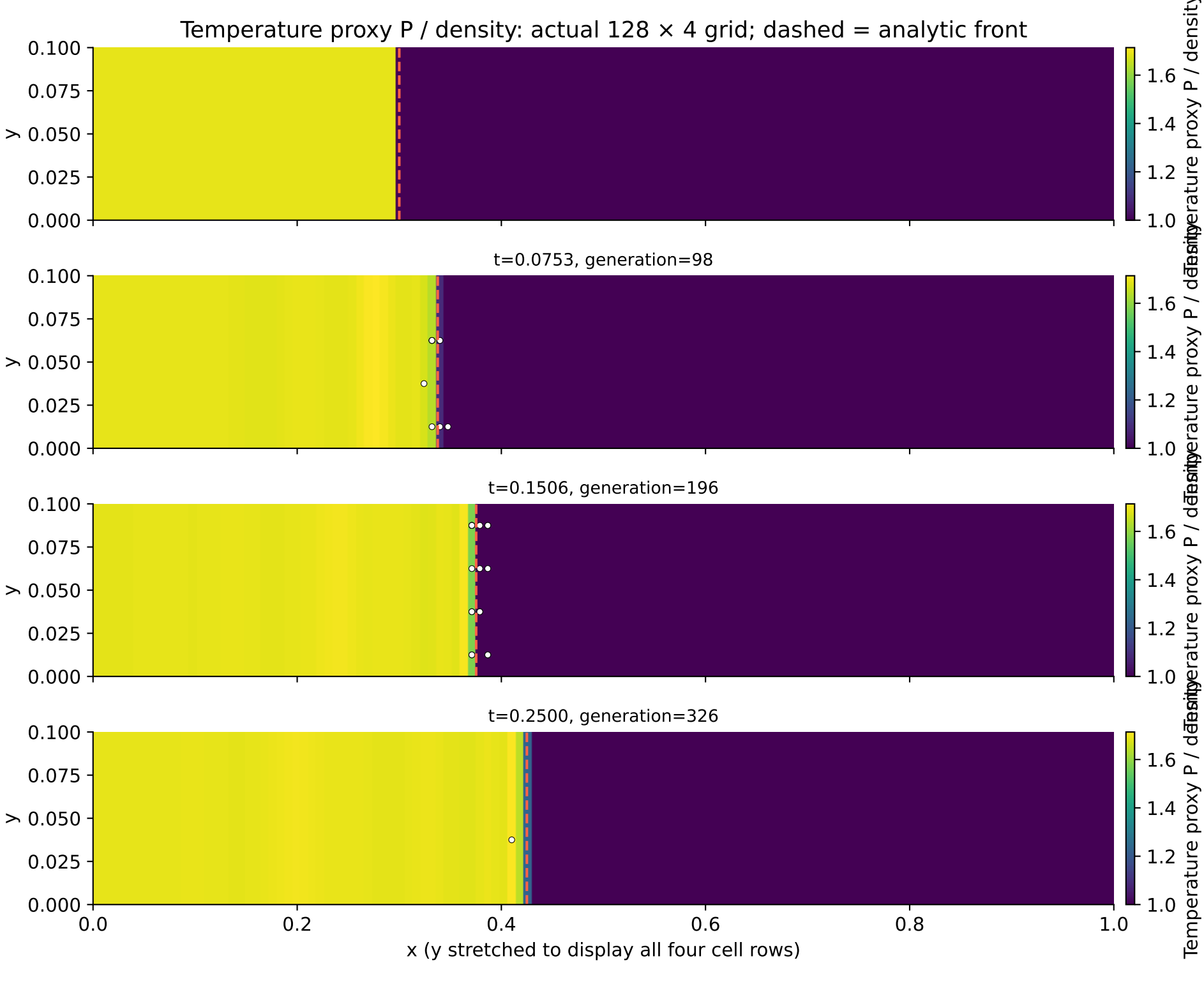
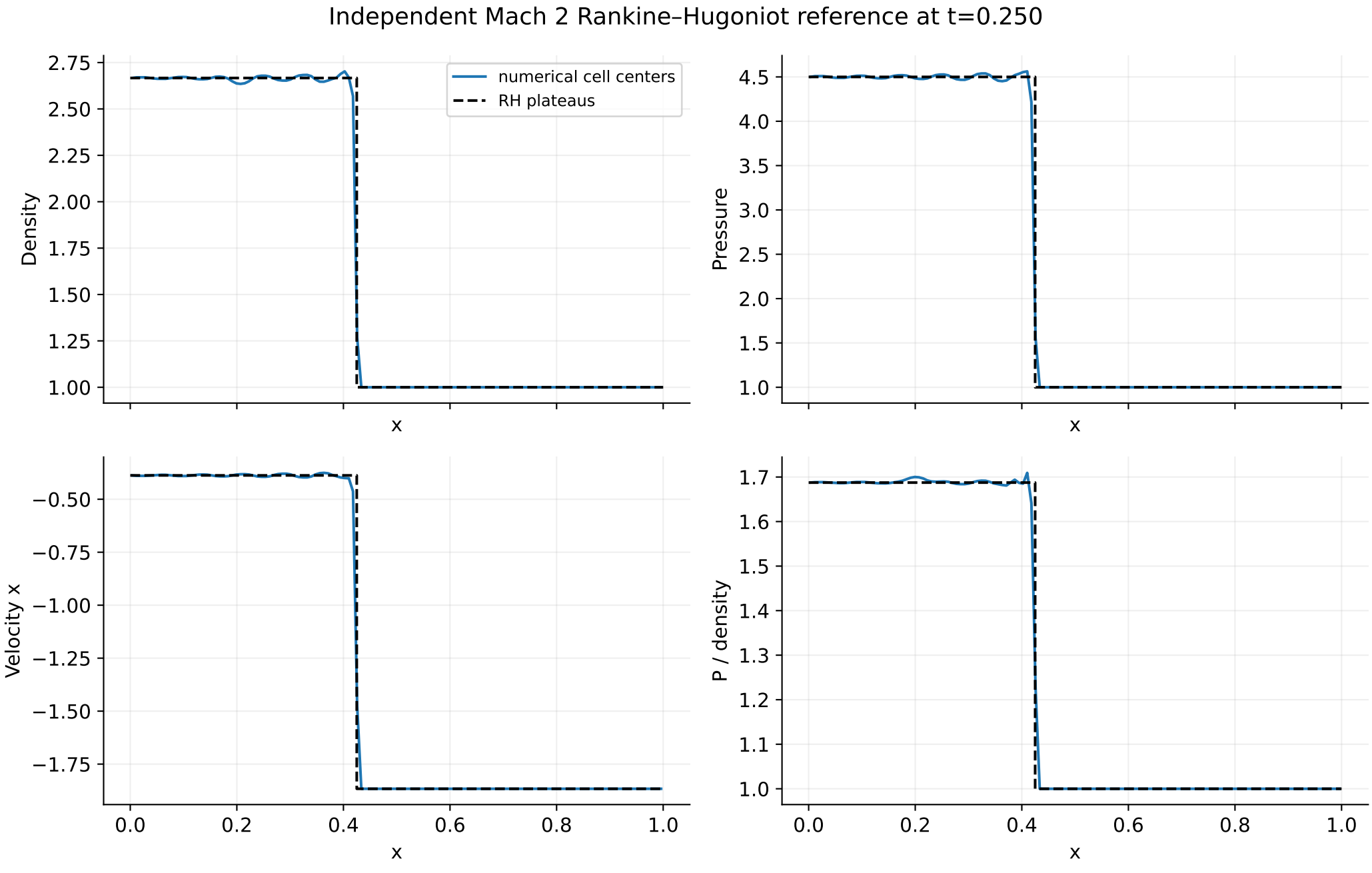
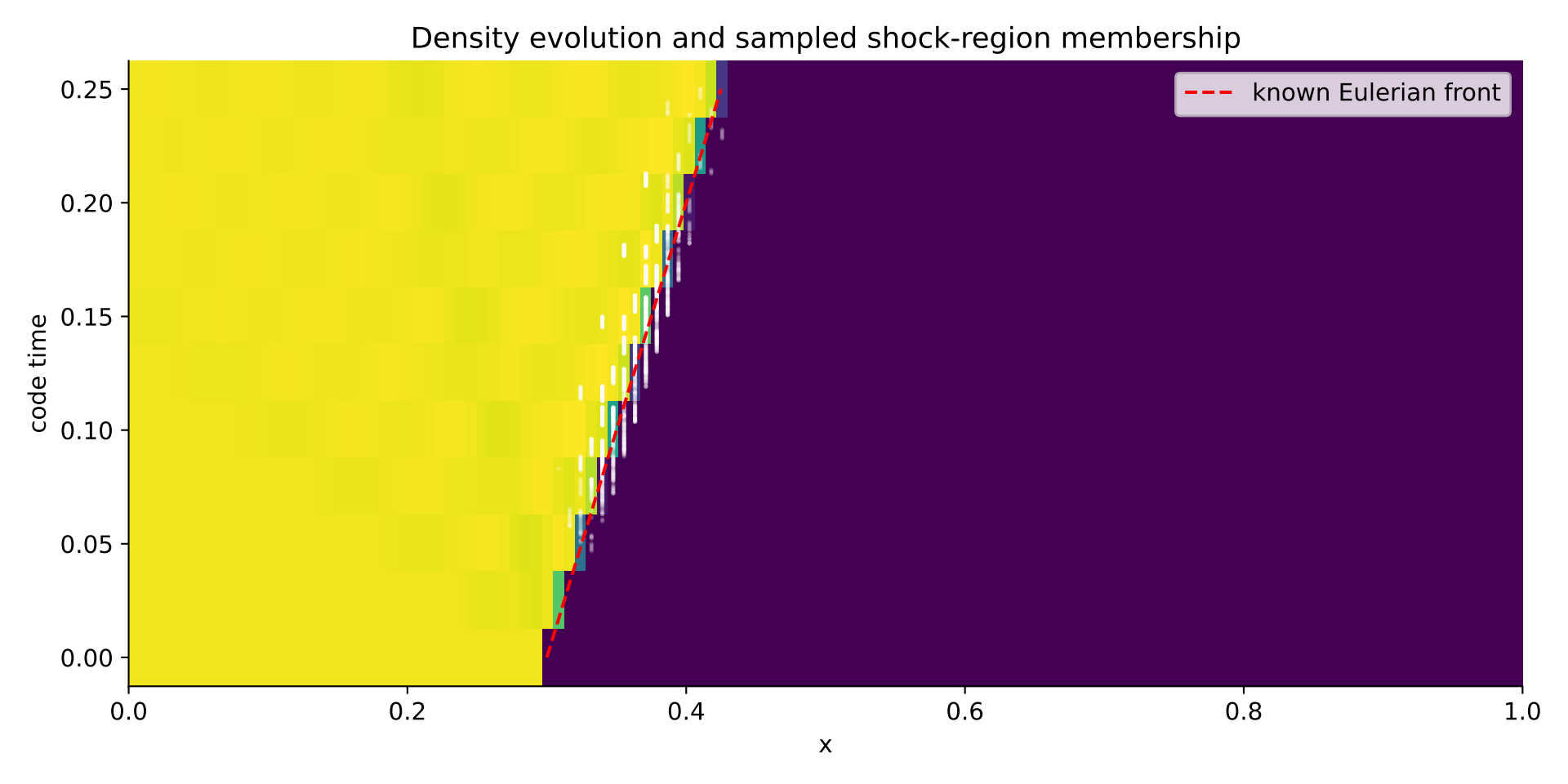
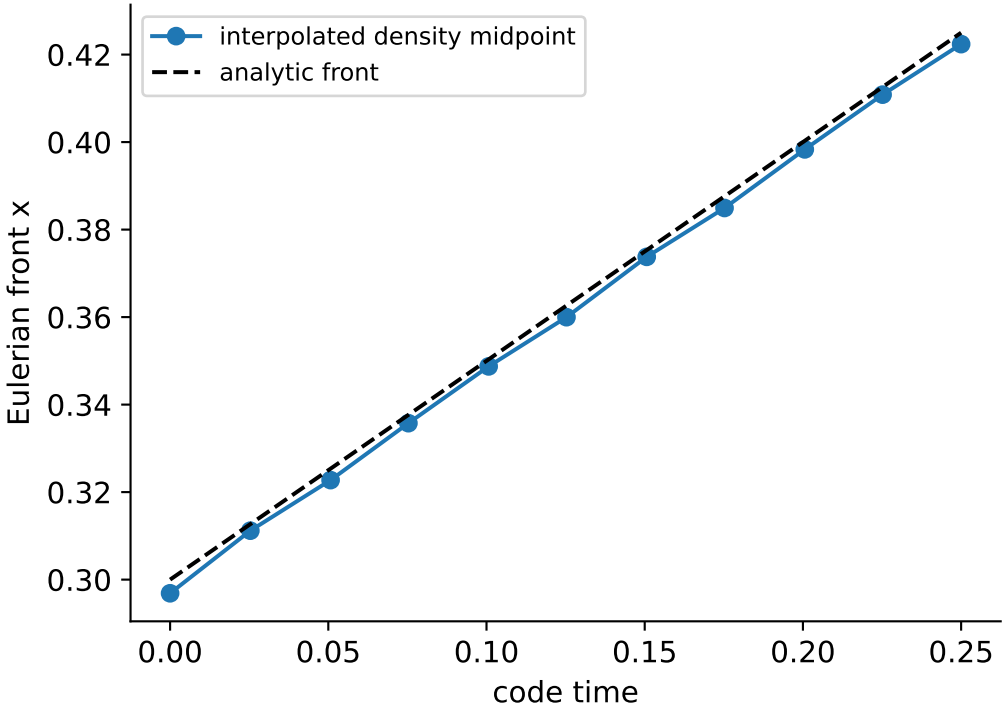

Density: actual 128 × 4 grid; dashed = analytic front
t=0.0753, generation=98
t=0.1506, generation=196
t=0.2500, generation=326
y
0.100
0.075
0.050
0.025
0.000
0.0
0.2
0.4
0.6
0.8
1.0
x (y stretched to display all four cell rows)
Density
2.5
2.0
1.5
1.0

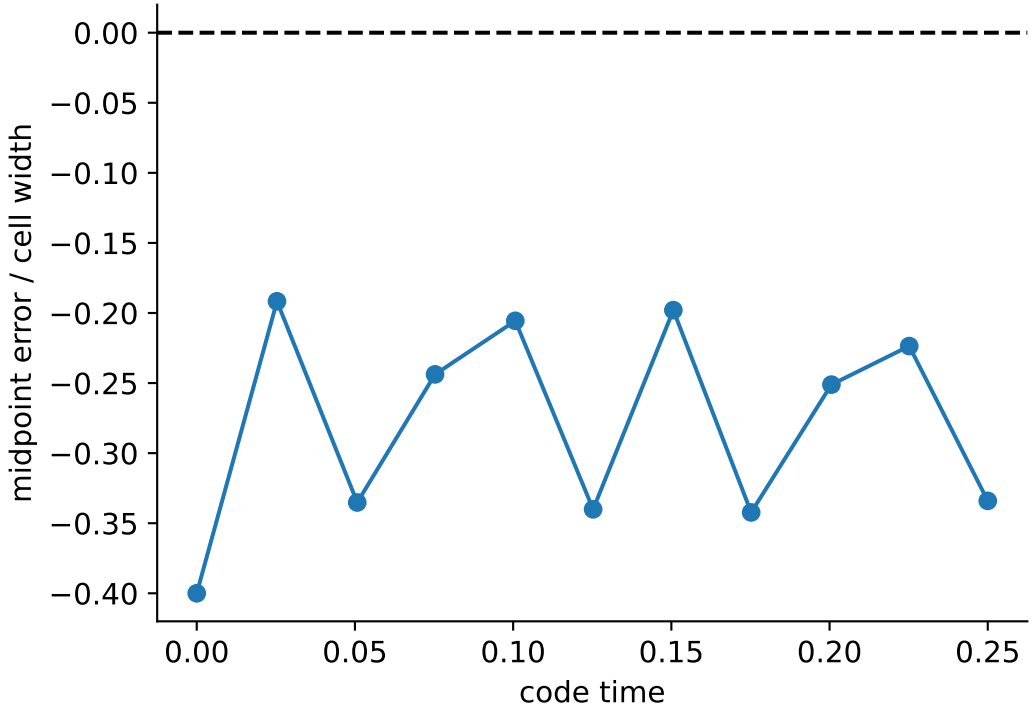

Pressure: actual 128 × 4 grid; dashed = analytic front
t=0.0753, generation=98
t=0.1506, generation=196
t=0.2500, generation=326
y
0.100
0.075
0.050
0.025
0.000
Pressure
1
2
3
4
0.0
0.2
0.4
0.6
0.8
1.0
x (y stretched to display all four cell rows)

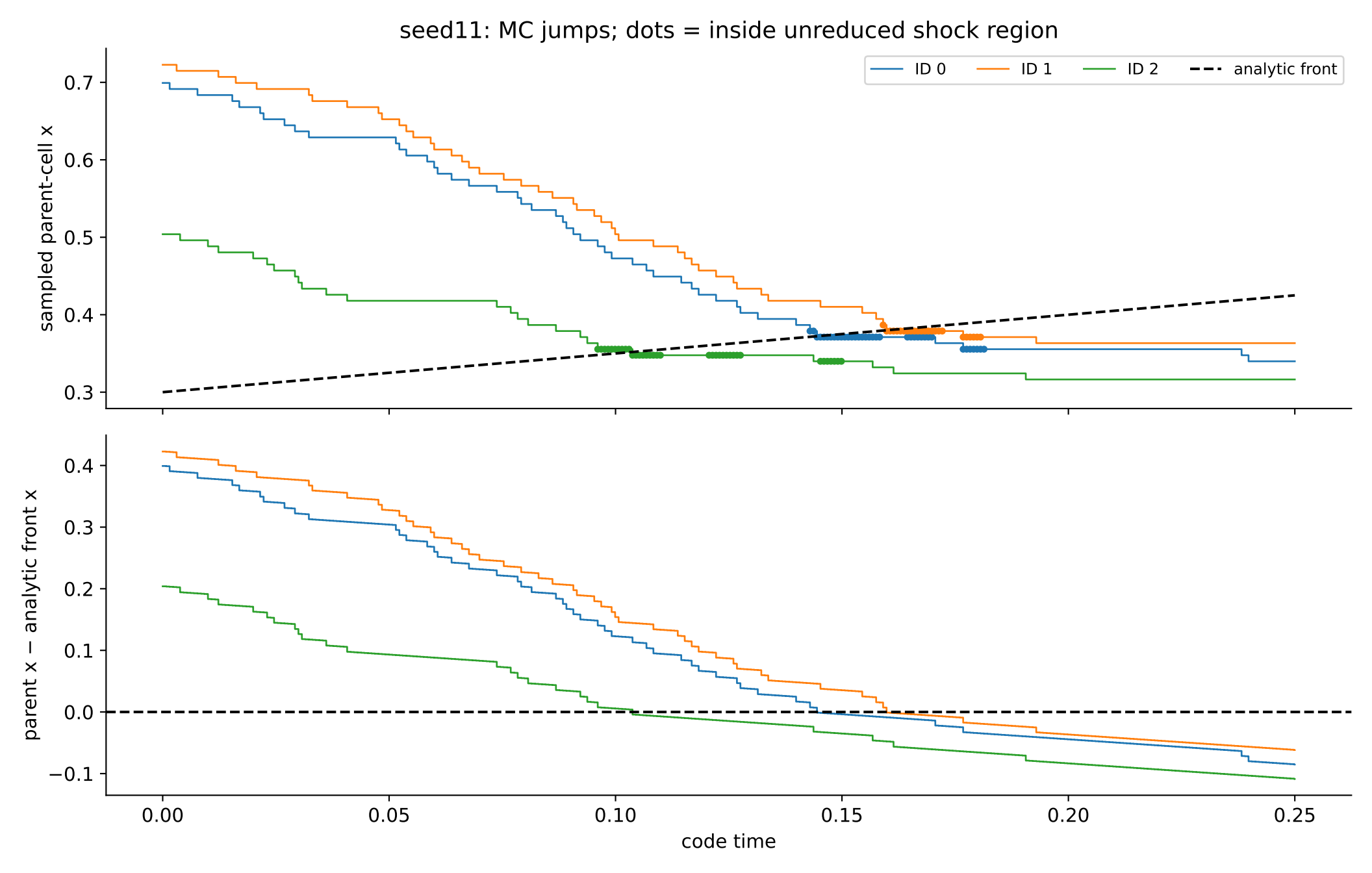

Temperature proxy P / density: actual 128 × 4 grid; dashed = analytic front
t=0.0753, generation=98
t=0.1506, generation=196
t=0.2500, generation=326
y
x (y stretched to display all four cell rows)
Temperature proxy P / density

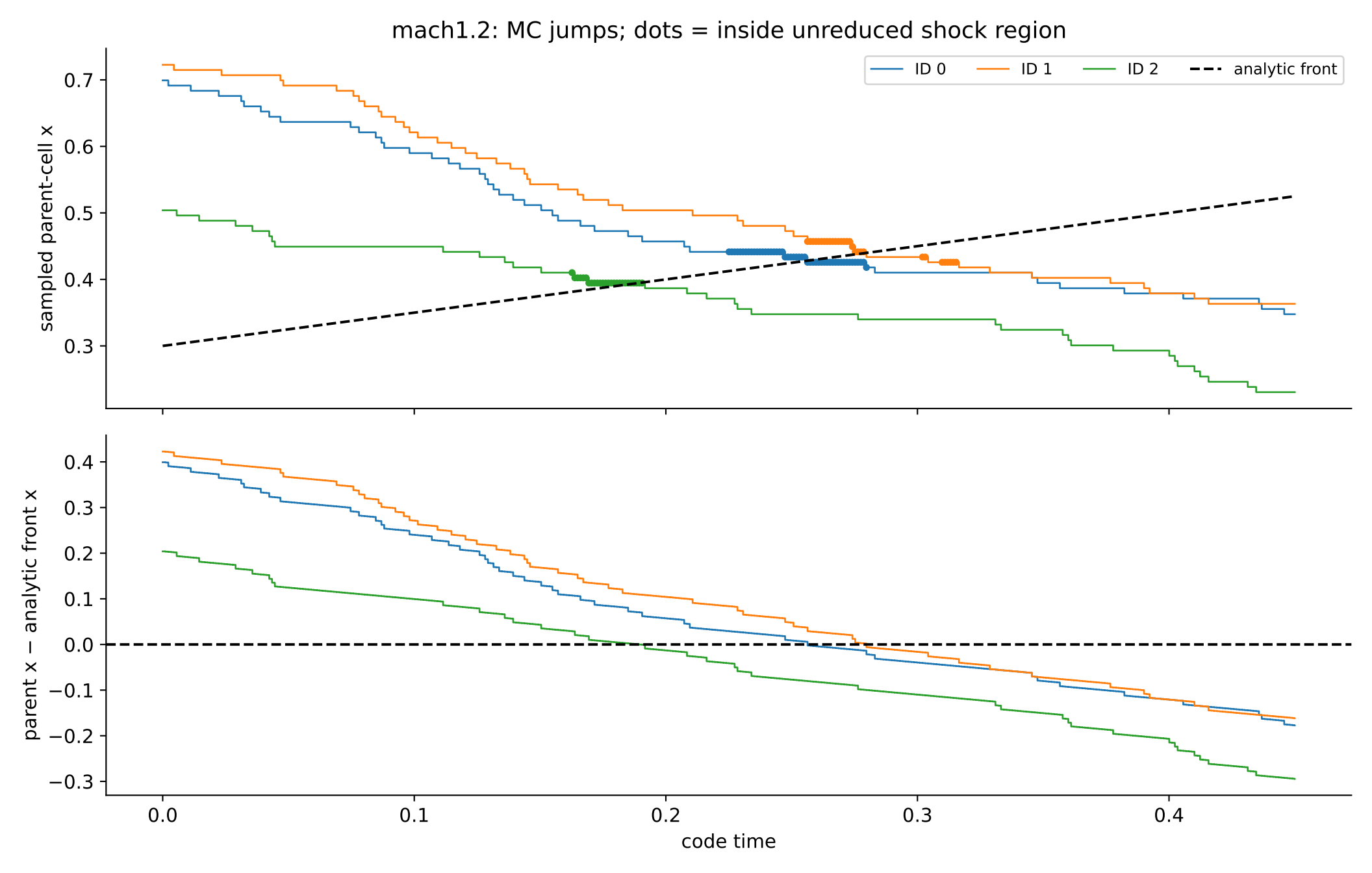

Independent Mach 2 Rankine–Hugoniot reference at t=0.250
numerical cell centers
RH plateaus
Density
Pressure
Velocity x
P / density
x
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
1.7
1.6
1.5
1.4
1.3
1.2
1.1
0.0
0.2
0.4
0.6
0.8
1.0

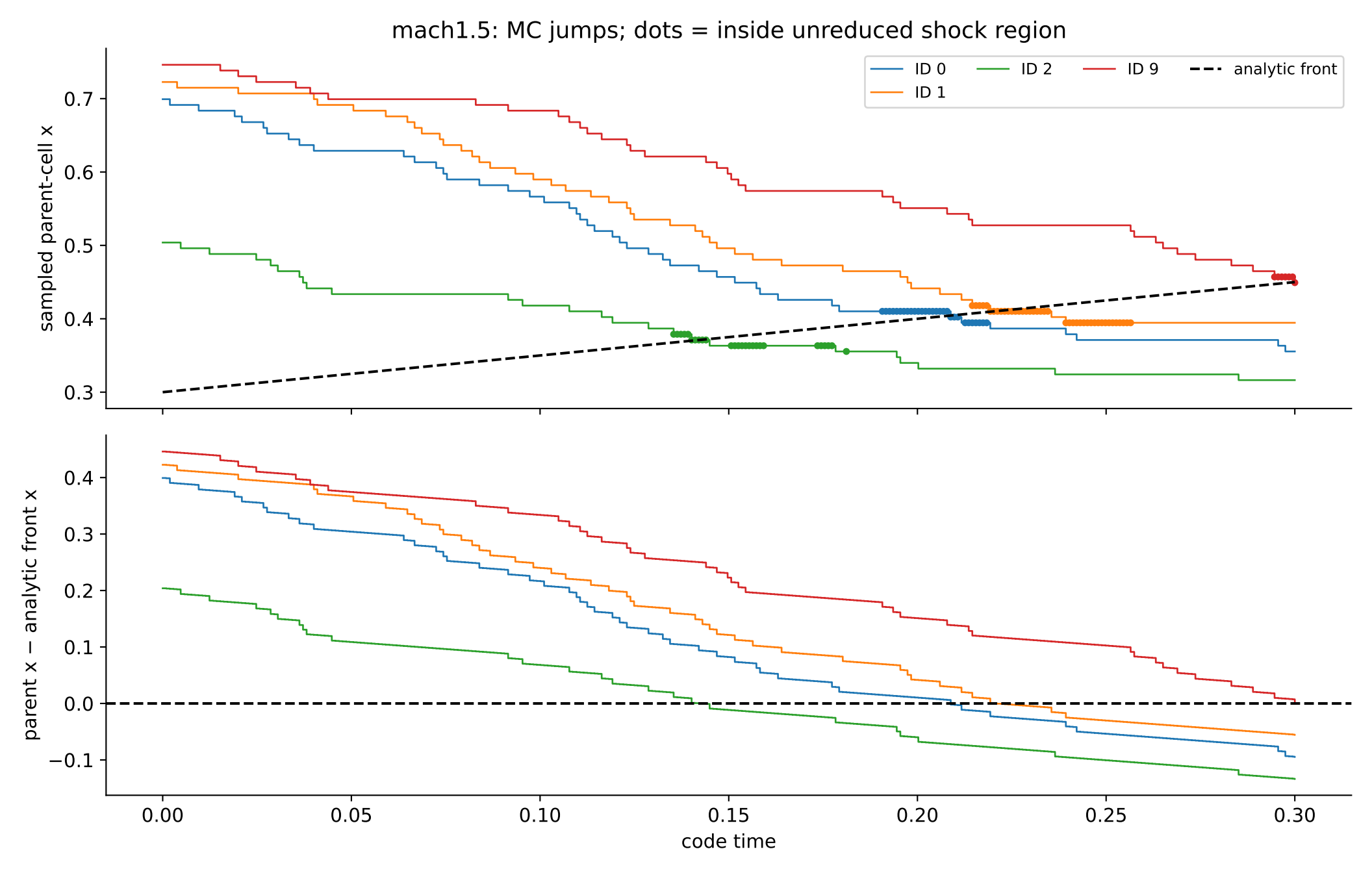

Density evolution and sampled shock-region membership
known Eulerian front
code time
0.25
0.20
0.15
0.10
0.05
0.00
0.0
0.2
0.4
0.6
0.8
1.0
x

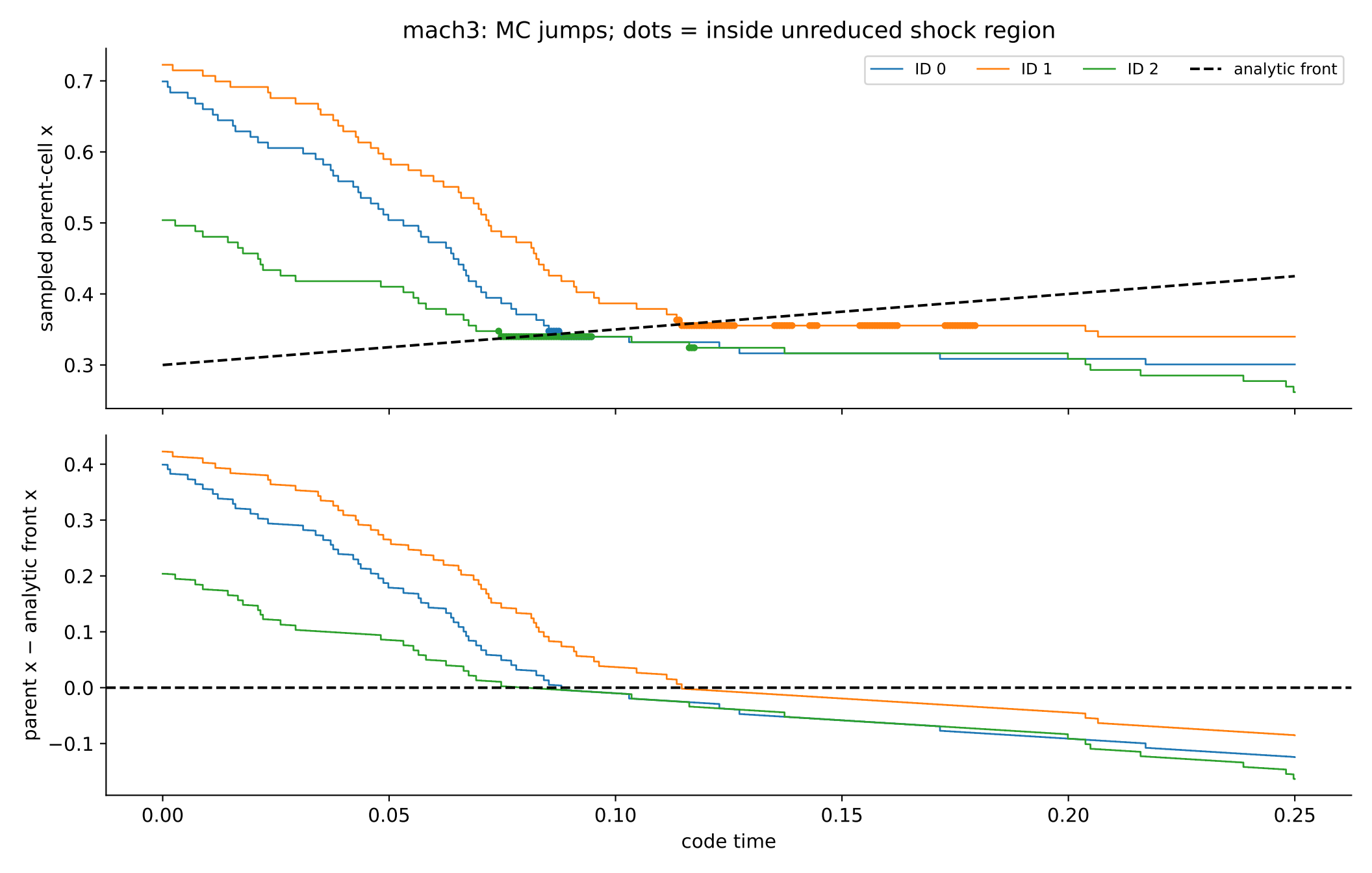

interpolated density midpoint
analytic front
Eulerian front x
0.42
0.40
0.38
0.36
0.34
0.32
0.30
0.00
0.05
0.10
0.15
0.20
0.25
code time

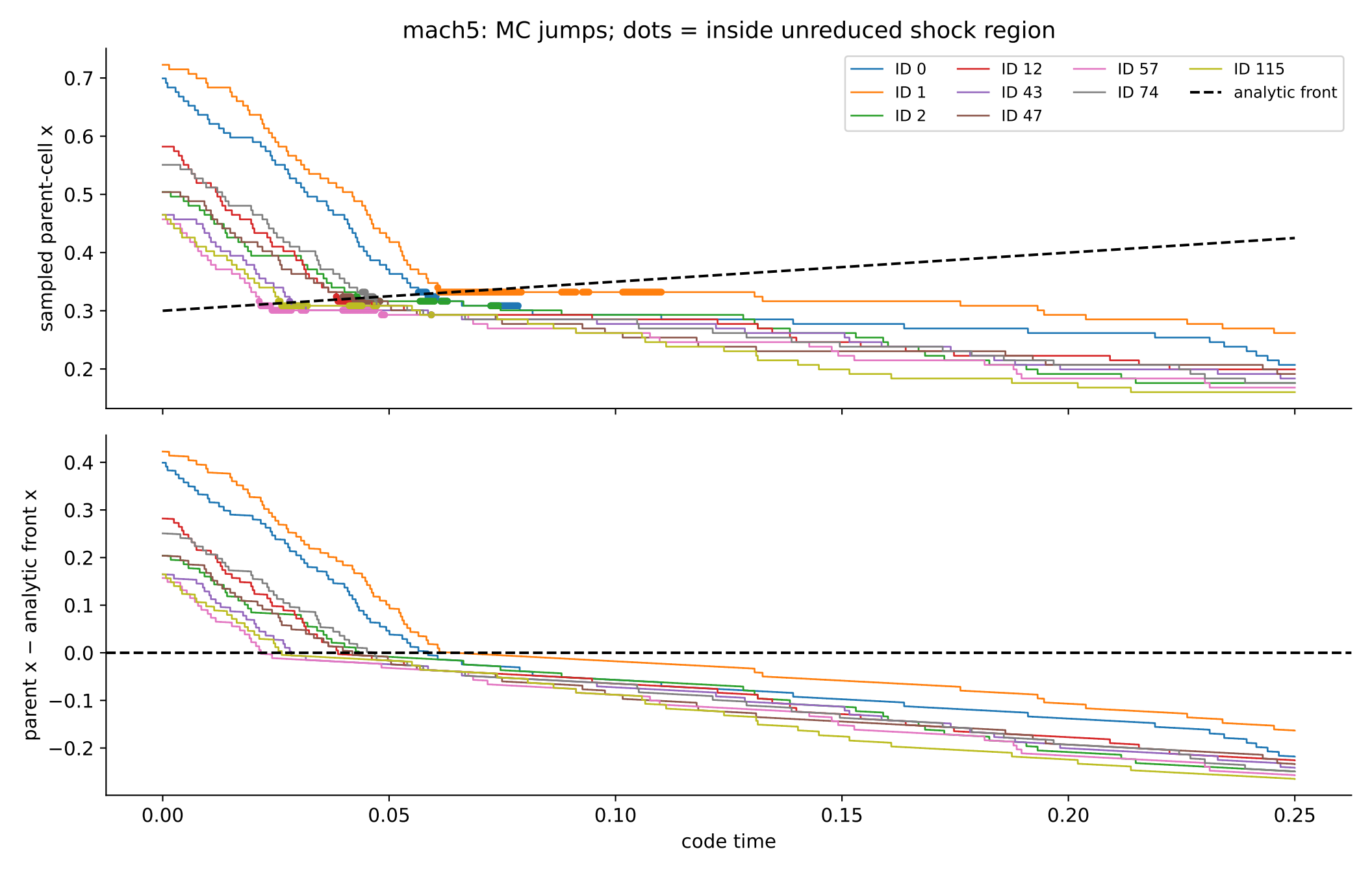

midpoint error / cell width
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
0.00
0.05
0.10
0.15
0.20
0.25
code time

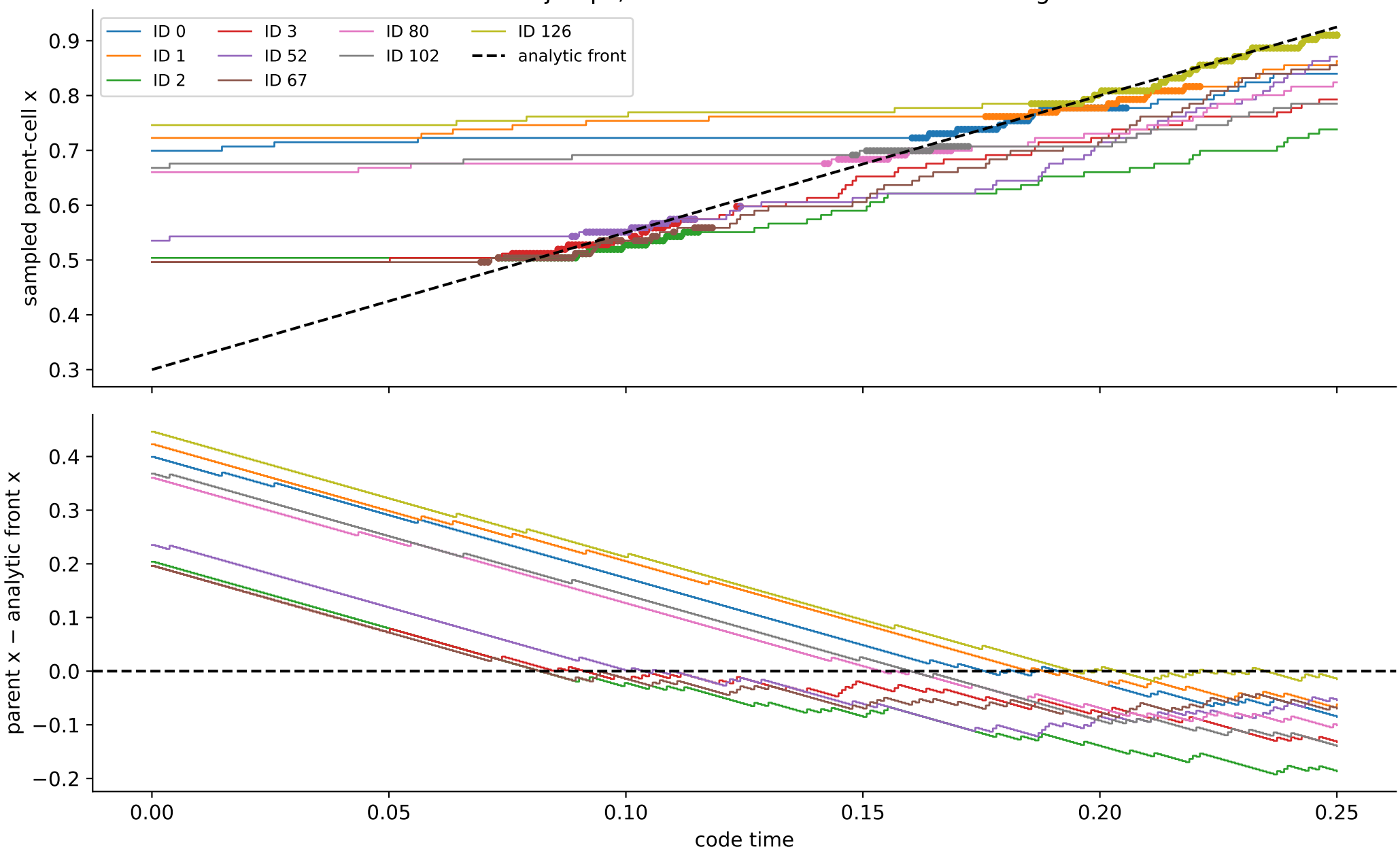

seed11: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
code time

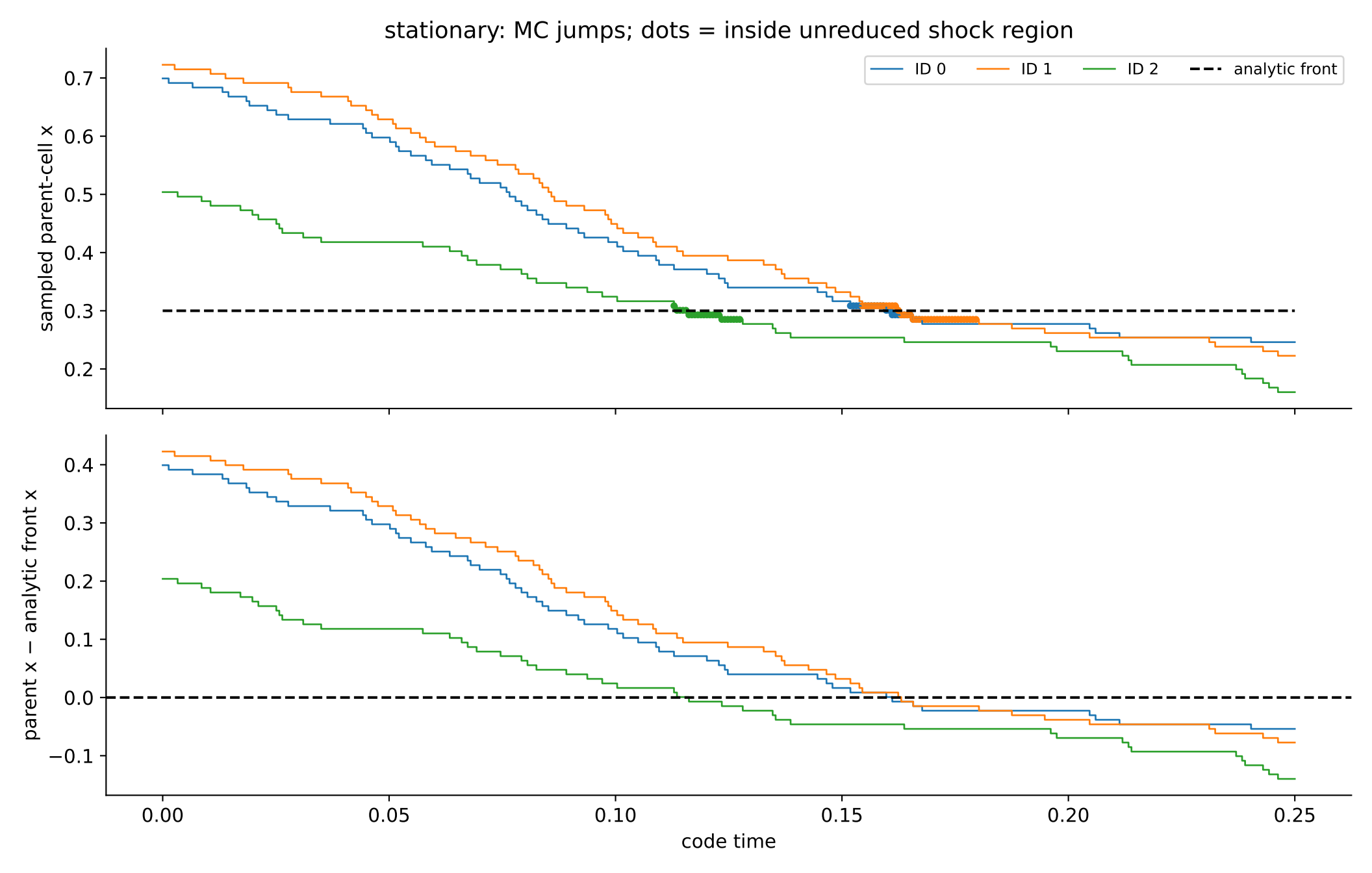

mach1.2: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0.0
0.1
0.2
0.3
0.4
code time

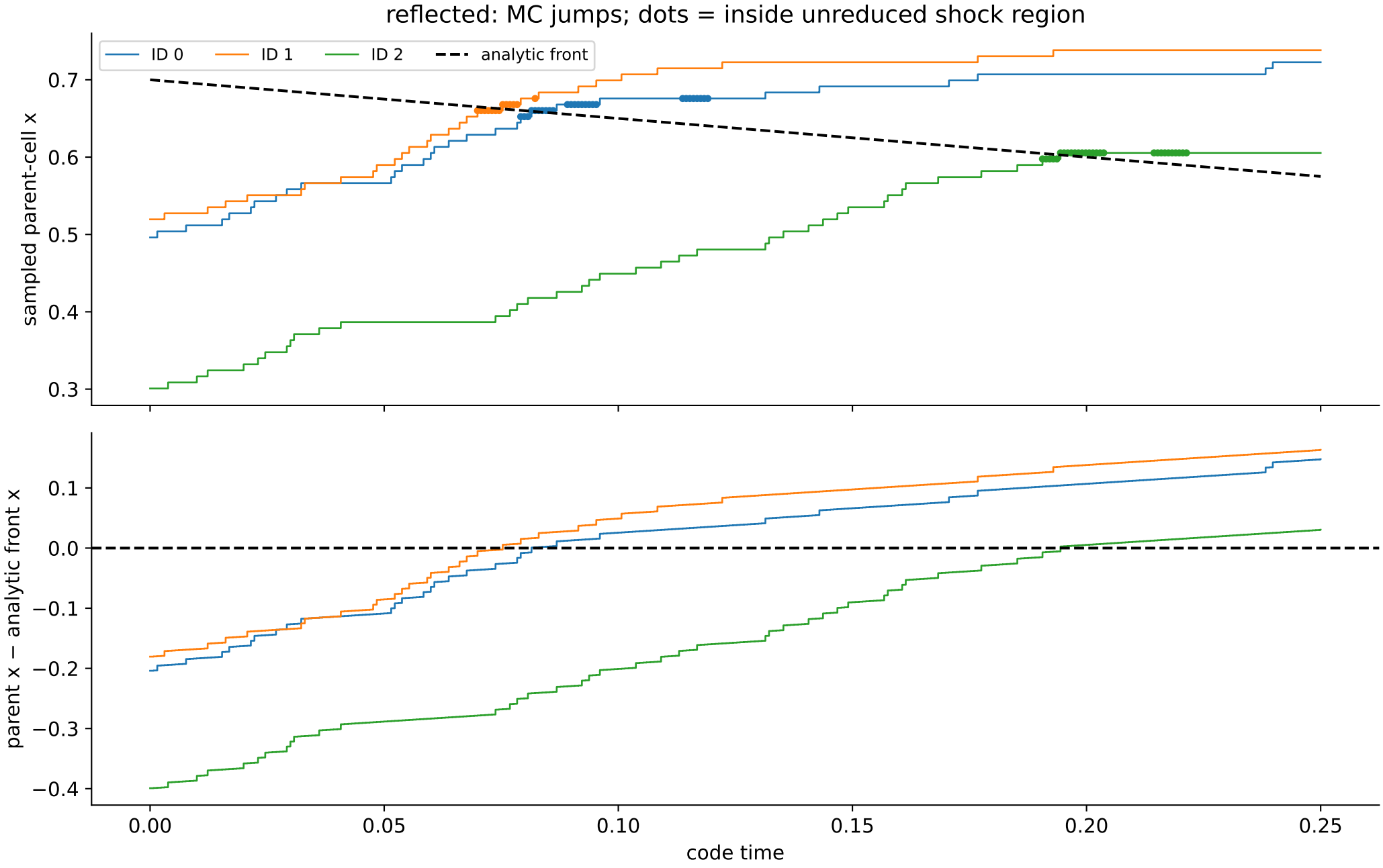

mach1.5: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
ID 9
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
0.30
code time

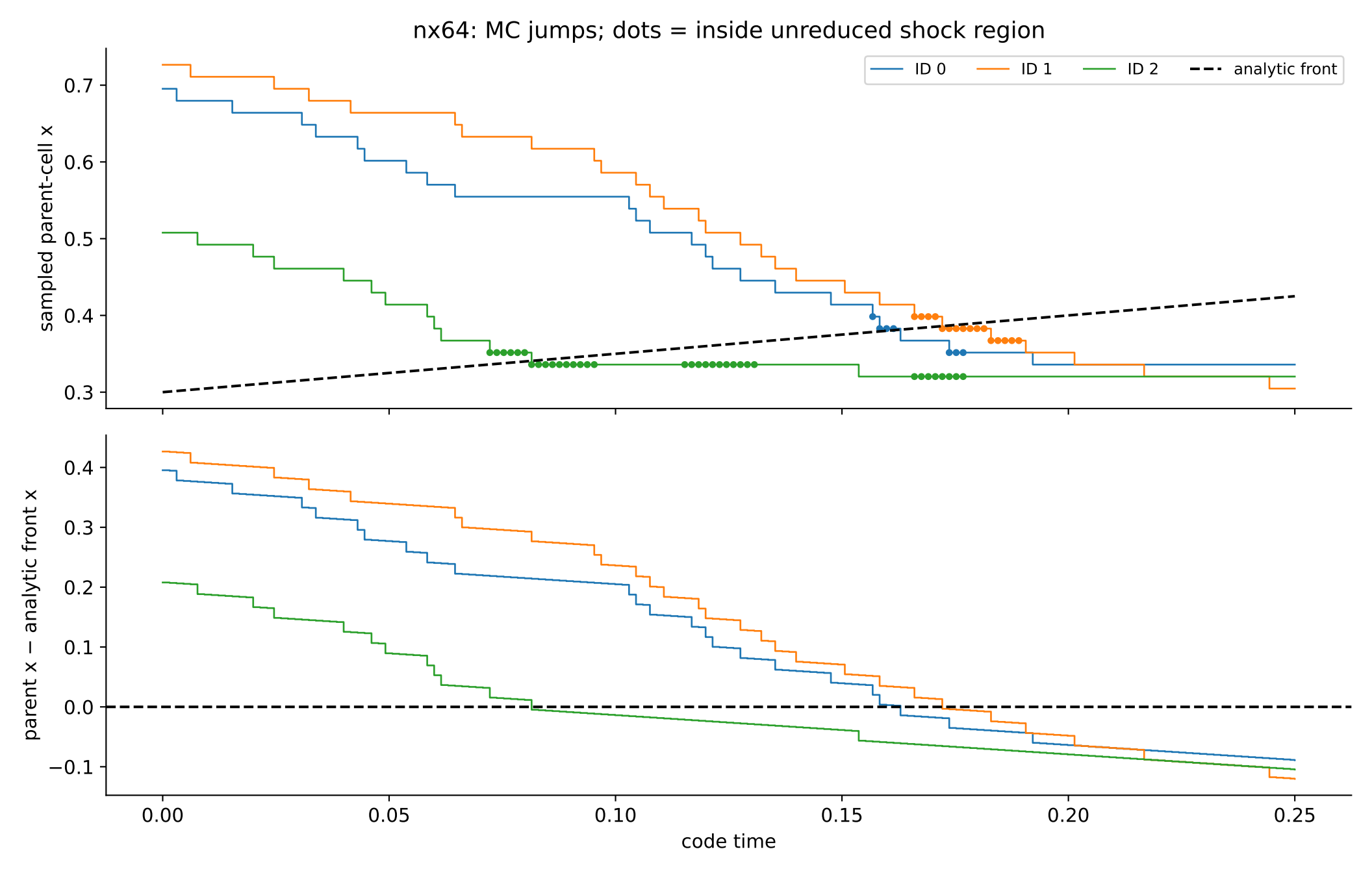

mach3: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
code time

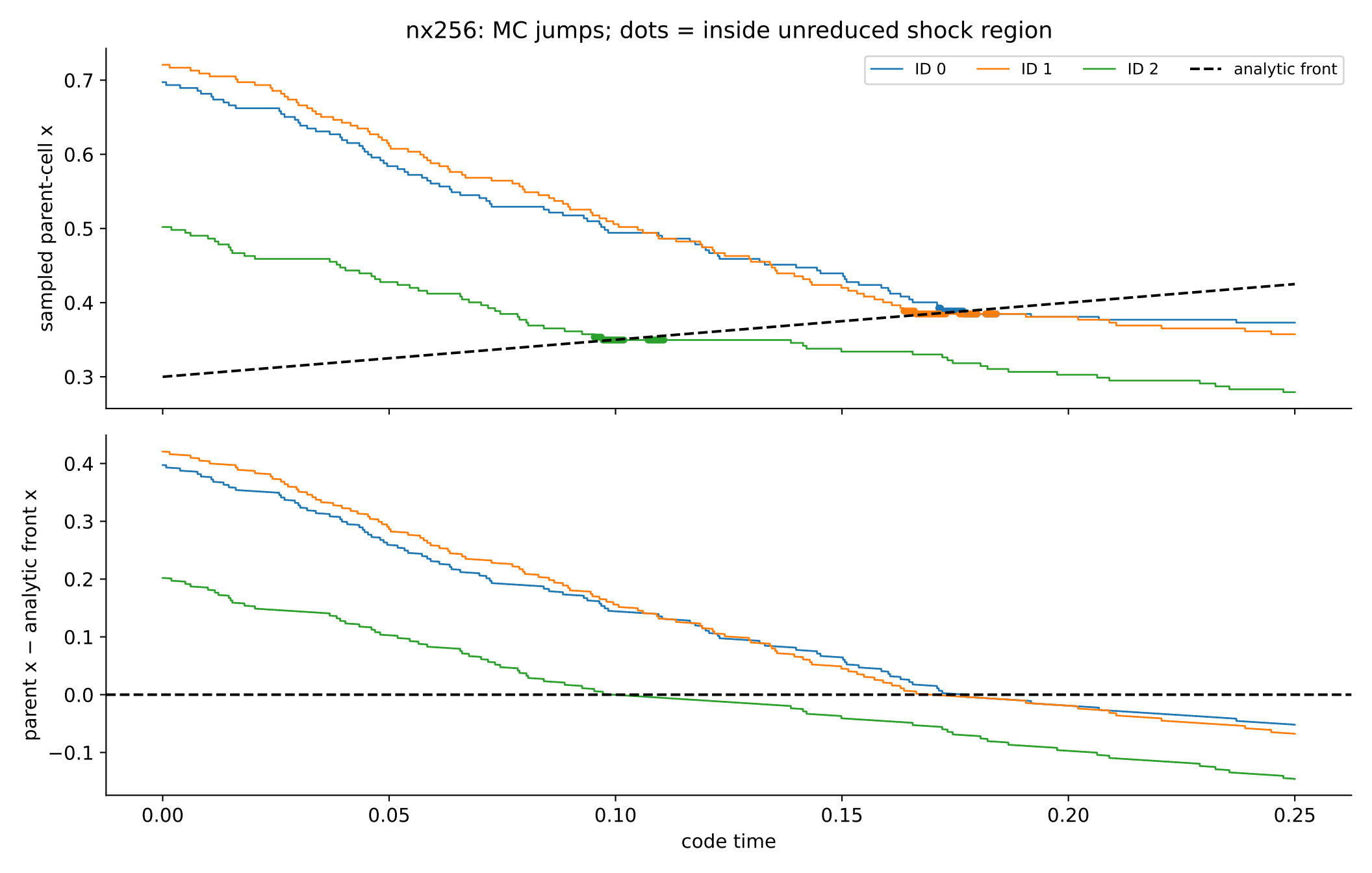

mach5: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
ID 12
ID 43
ID 47
ID 57
ID 74
ID 115
analytic front
sampled parent-cell x
parent x − analytic front x
code time
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0.00
0.05
0.10
0.15
0.20
0.25

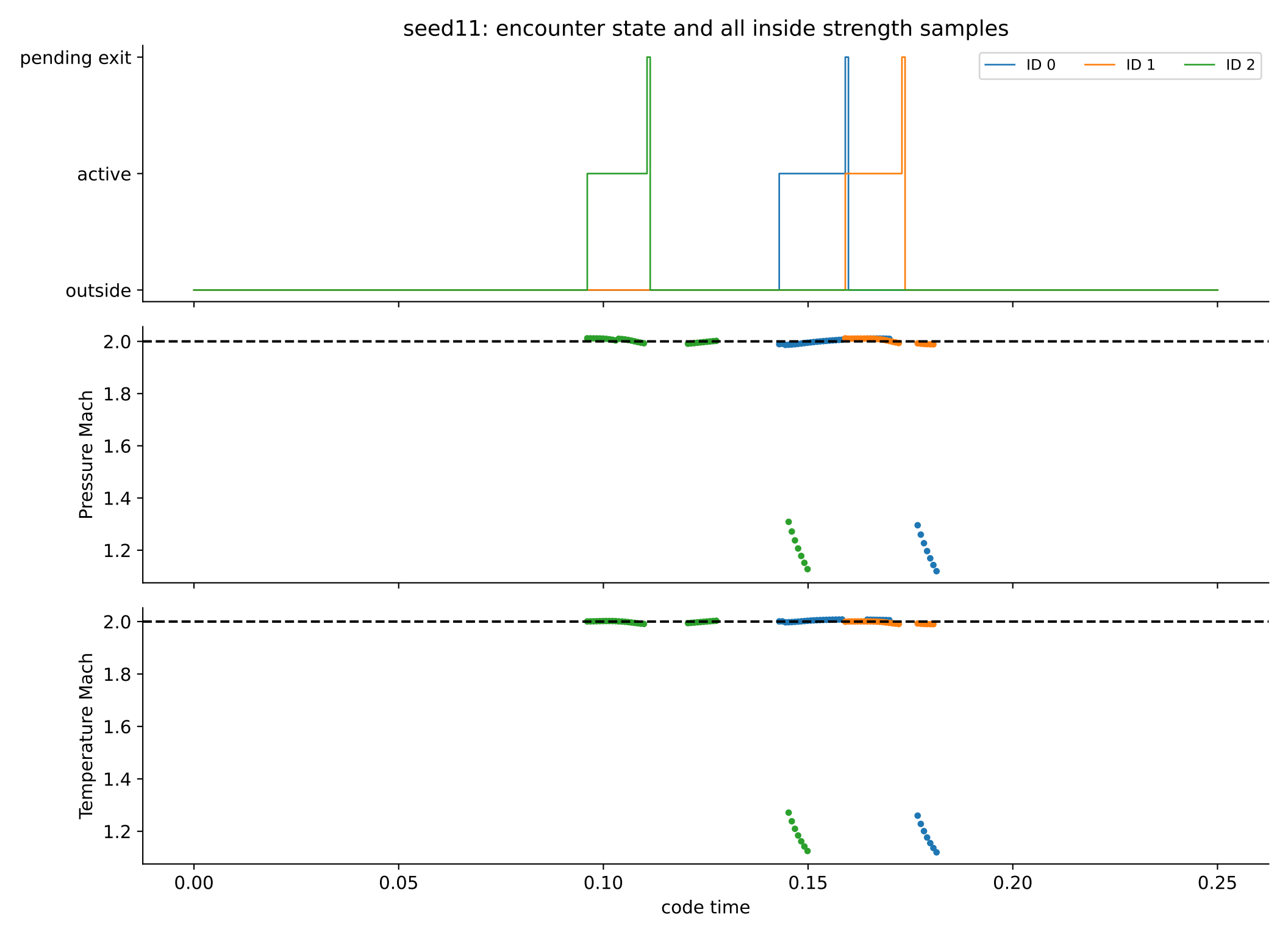

boost2: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
ID 3
ID 52
ID 67
ID 80
ID 102
ID 126
analytic front
sampled parent-cell x
parent x − analytic front x
code time
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.00
0.05
0.10
0.15
0.20
0.25

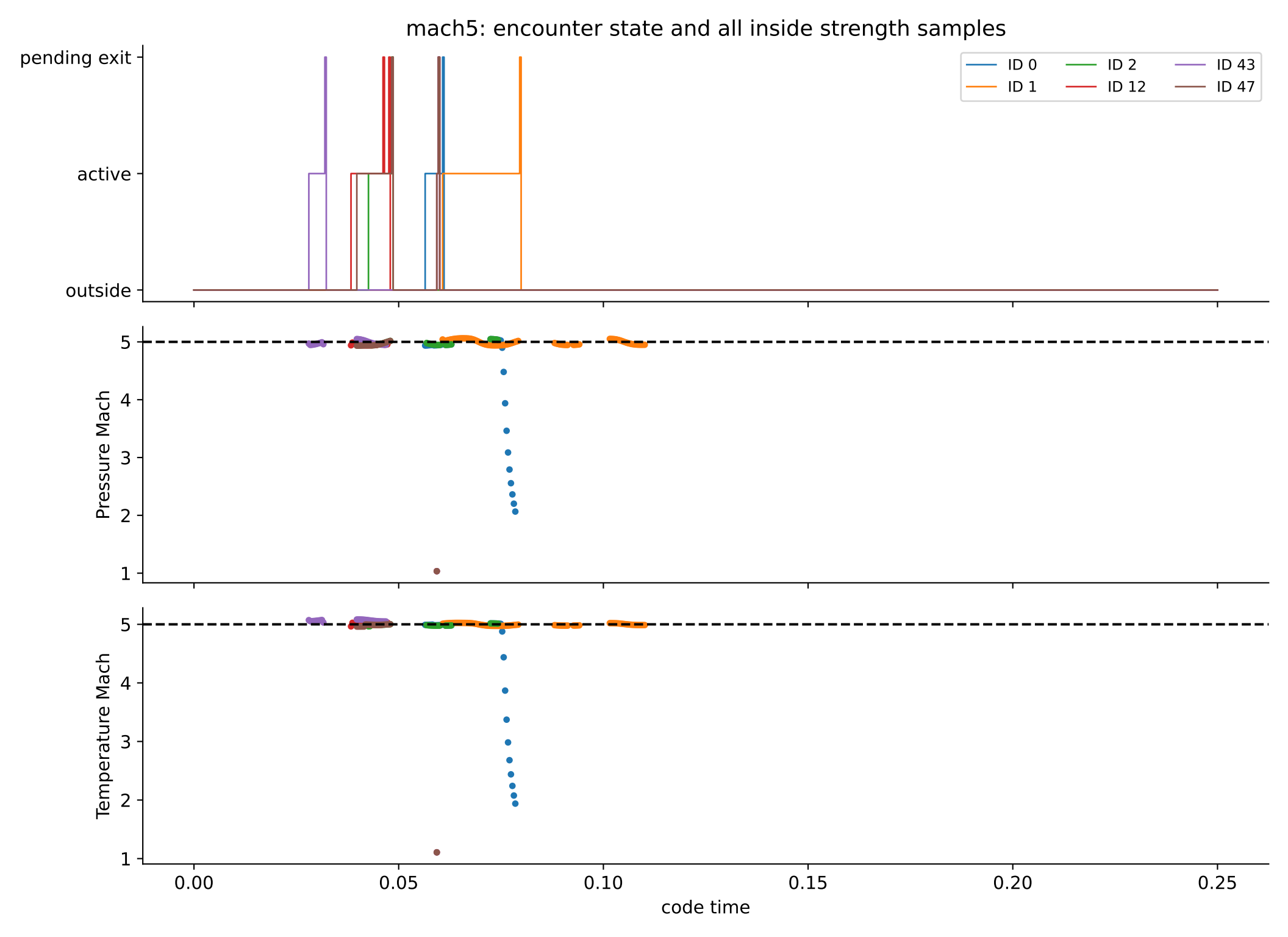

stationary: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
0.2
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
code time

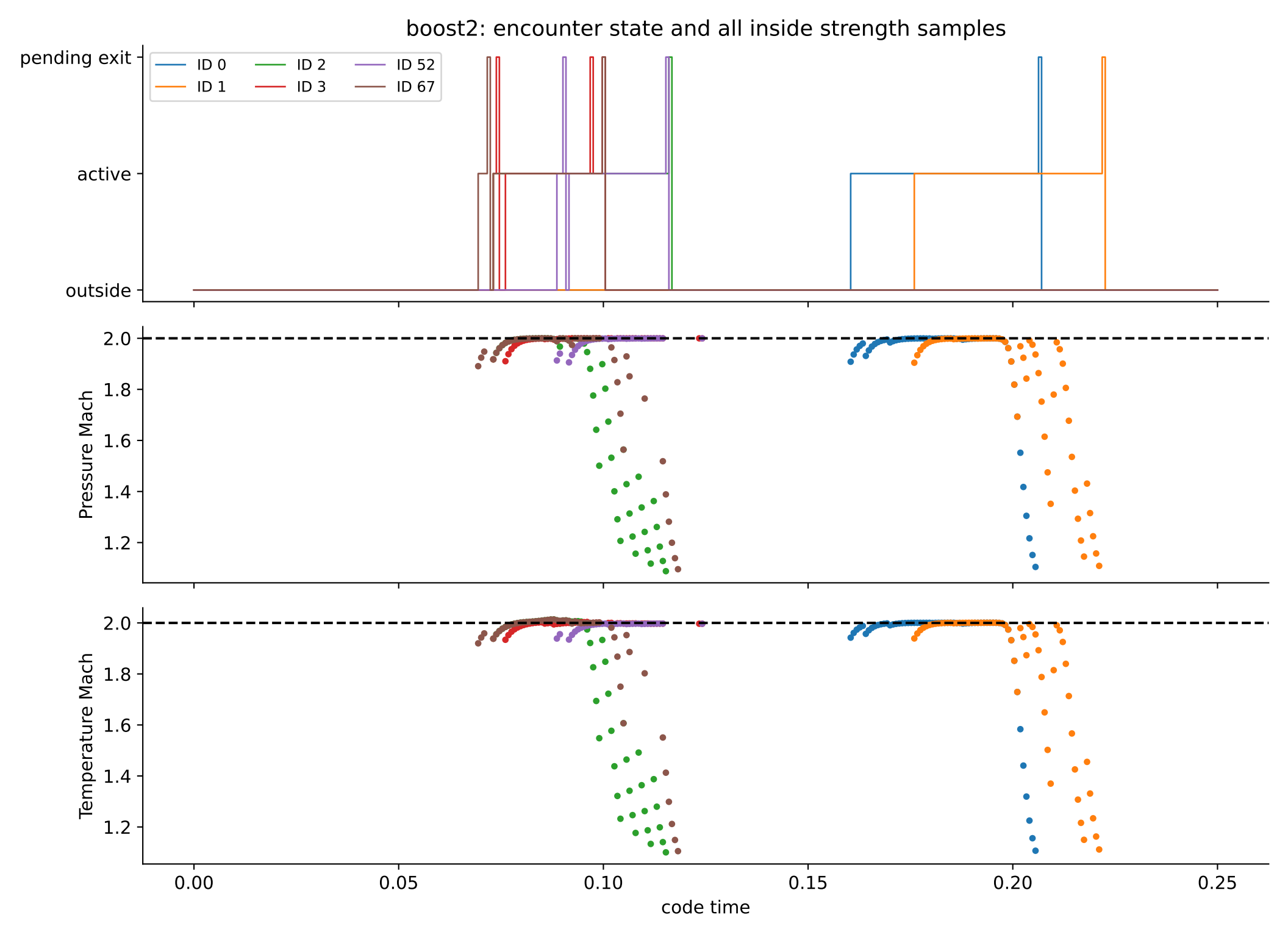

reflected: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0.00
0.05
0.10
0.15
0.20
0.25
code time

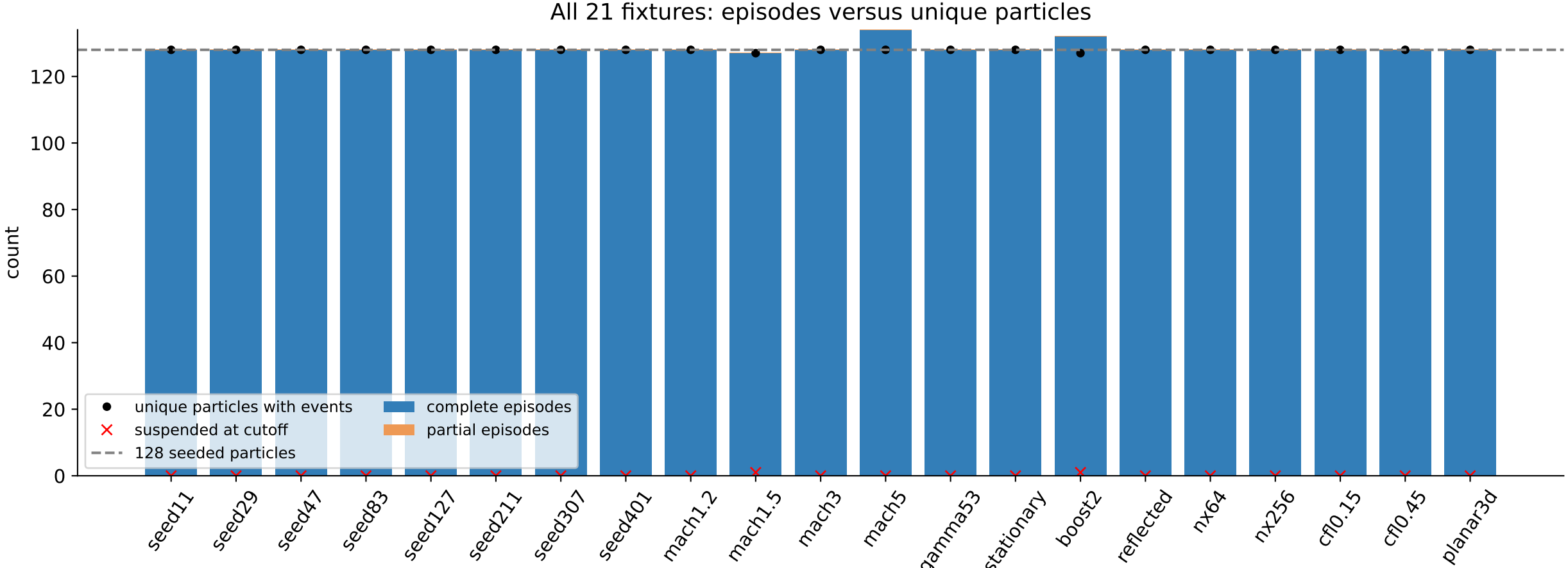

nx64: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
code time

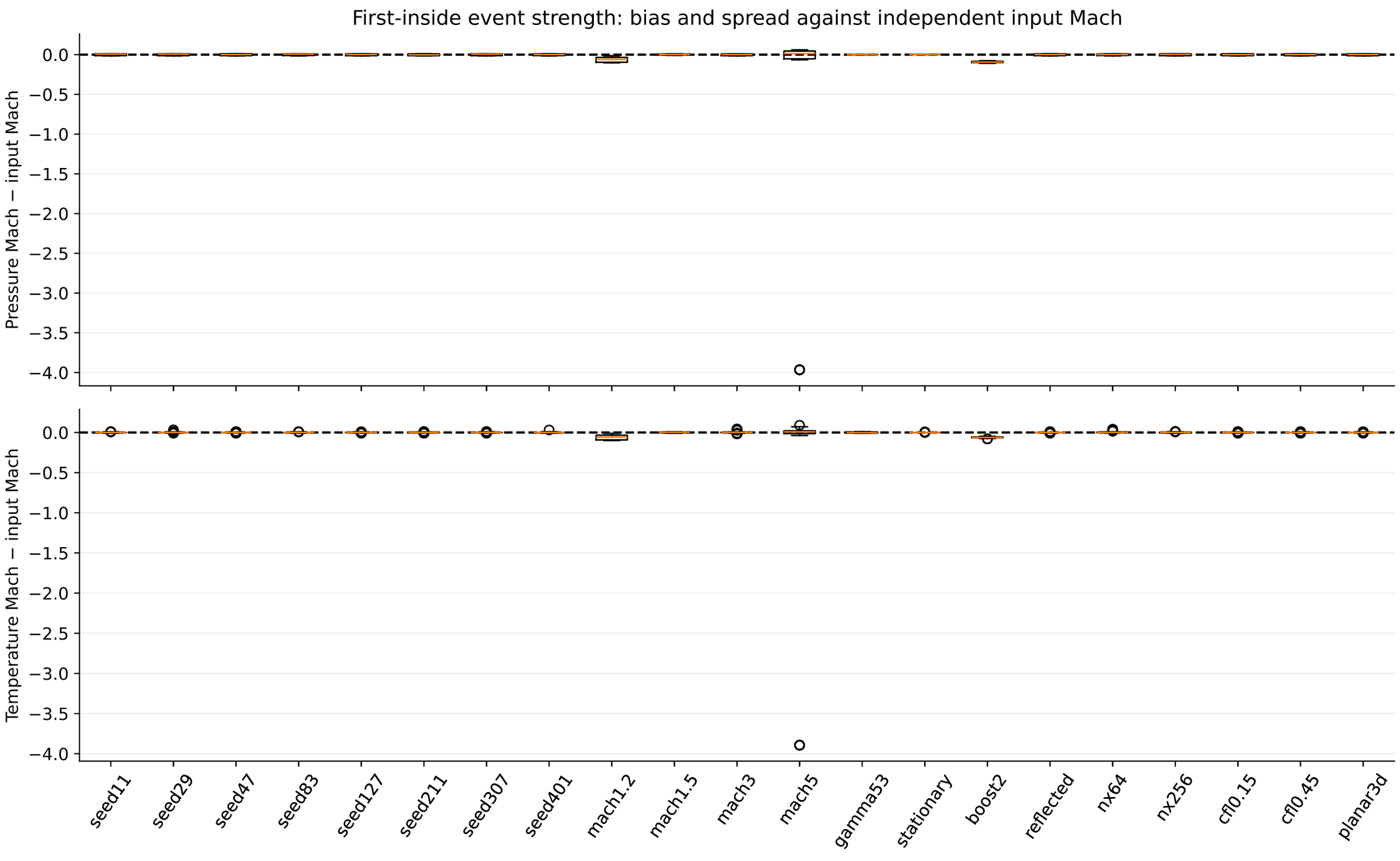

nx256: MC jumps; dots = inside unreduced shock region
ID 0
ID 1
ID 2
analytic front
sampled parent-cell x
0.7
0.6
0.5
0.4
0.3
parent x − analytic front x
0.4
0.3
0.2
0.1
0.0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
code time

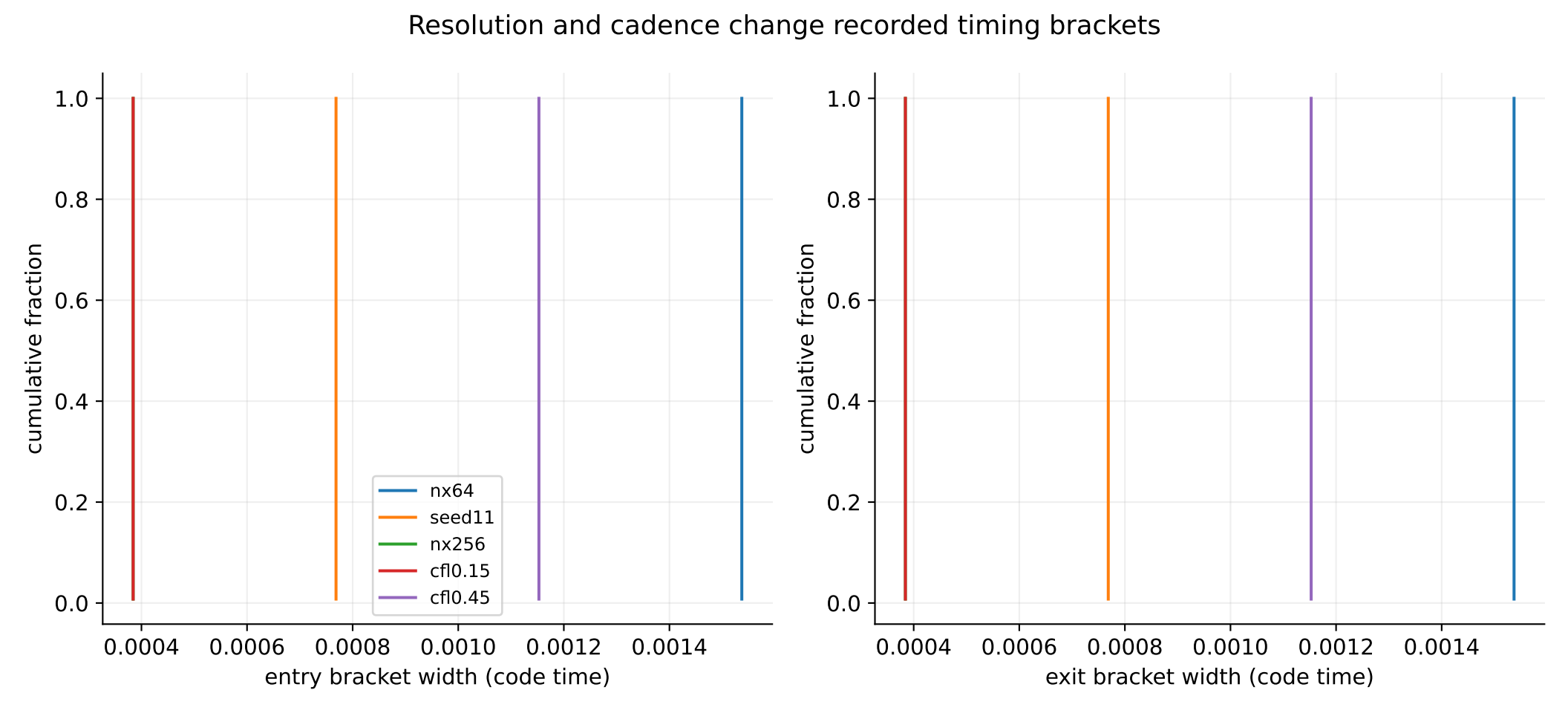

seed11: encounter state and all inside strength samples
pending exit
active
outside
ID 0
ID 1
ID 2
Pressure Mach
2.0
1.8
1.6
1.4
1.2
Temperature Mach
2.0
1.8
1.6
1.4
1.2
0.00
0.05
0.10
0.15
0.20
0.25
code time

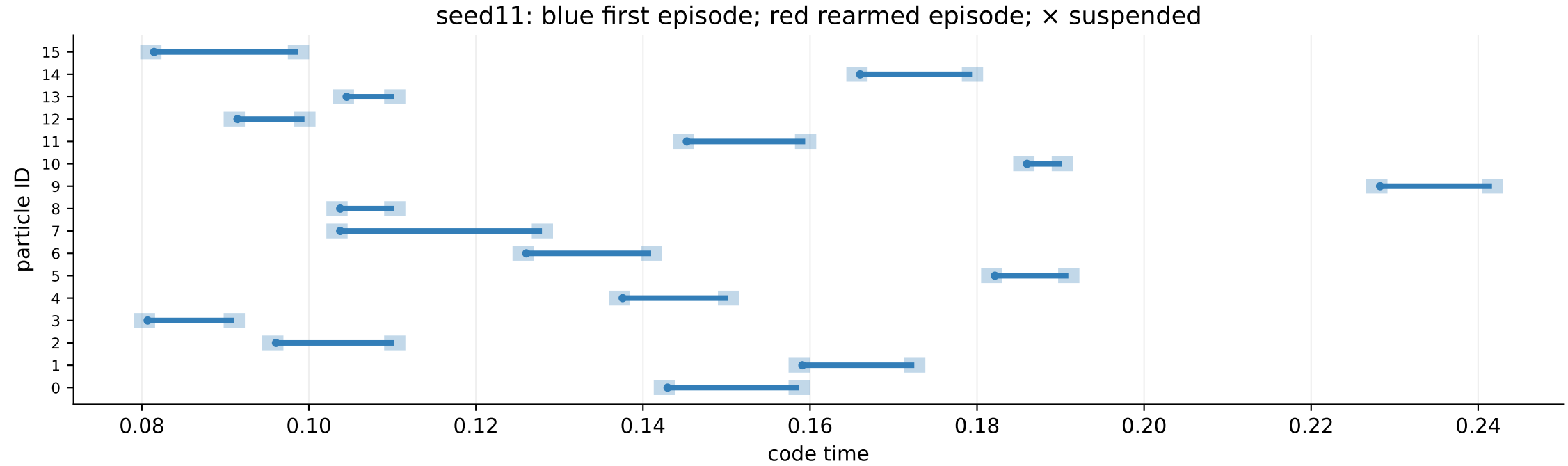

mach5: encounter state and all inside strength samples
pending exit
active
outside
ID 0
ID 1
ID 2
ID 12
ID 43
ID 47
Pressure Mach
Temperature Mach
1
2
3
4
5
0.00
0.05
0.10
0.15
0.20
0.25
code time

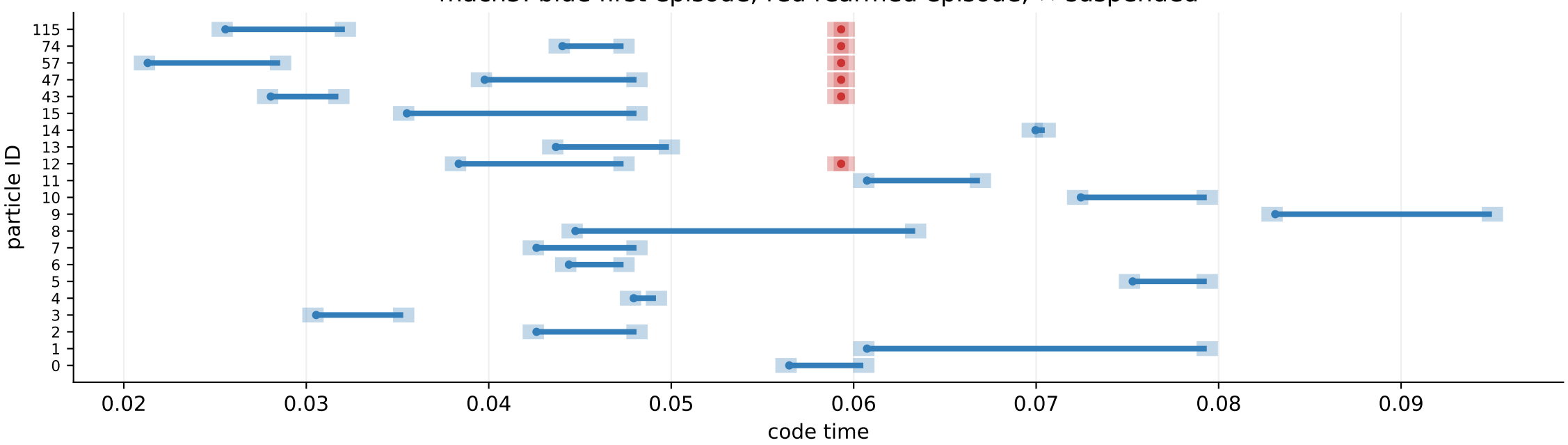

boost2: encounter state and all inside strength samples
pending exit
active
outside
ID 0
ID 1
ID 2
ID 3
ID 52
ID 67
Pressure Mach
Temperature Mach
2.0
1.8
1.6
1.4
1.2
0.00
0.05
0.10
0.15
0.20
0.25
code time

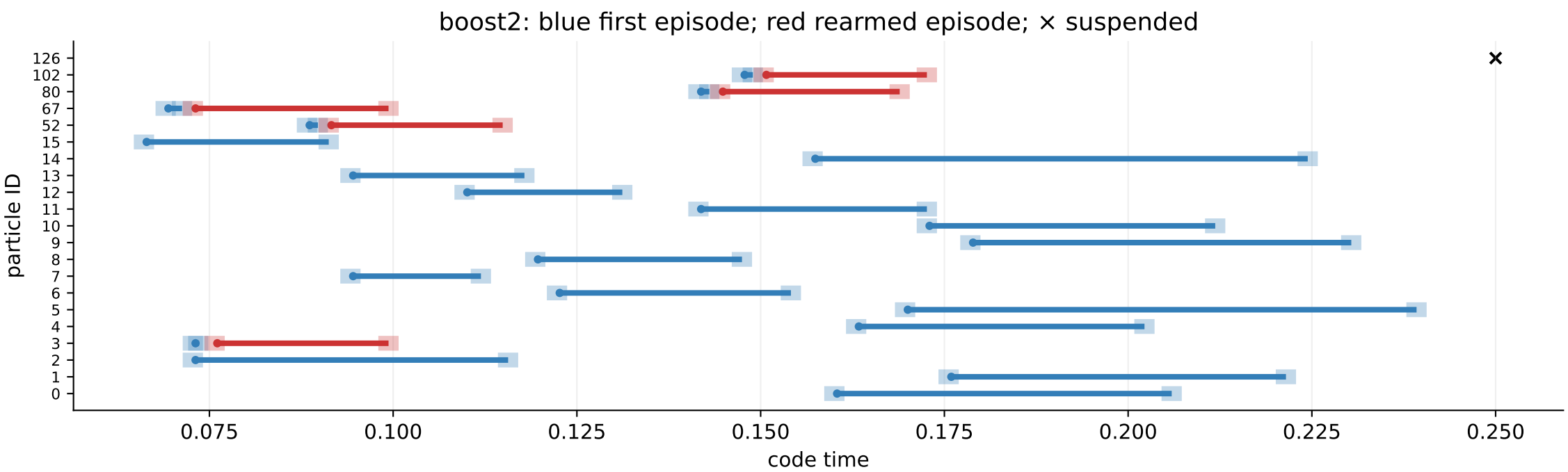

All 21 fixtures: episodes versus unique particles
count
0
20
40
60
80
100
120
unique particles with events
suspended at cutoff
128 seeded particles
complete episodes
partial episodes
seed11
seed29
seed47
seed83
seed127
seed211
seed307
seed401
mach1.2
mach1.5
mach3
mach5
gamma53
stationary
boost2
reflected
nx64
nx256
cfl0.15
cfl0.45
planar3d

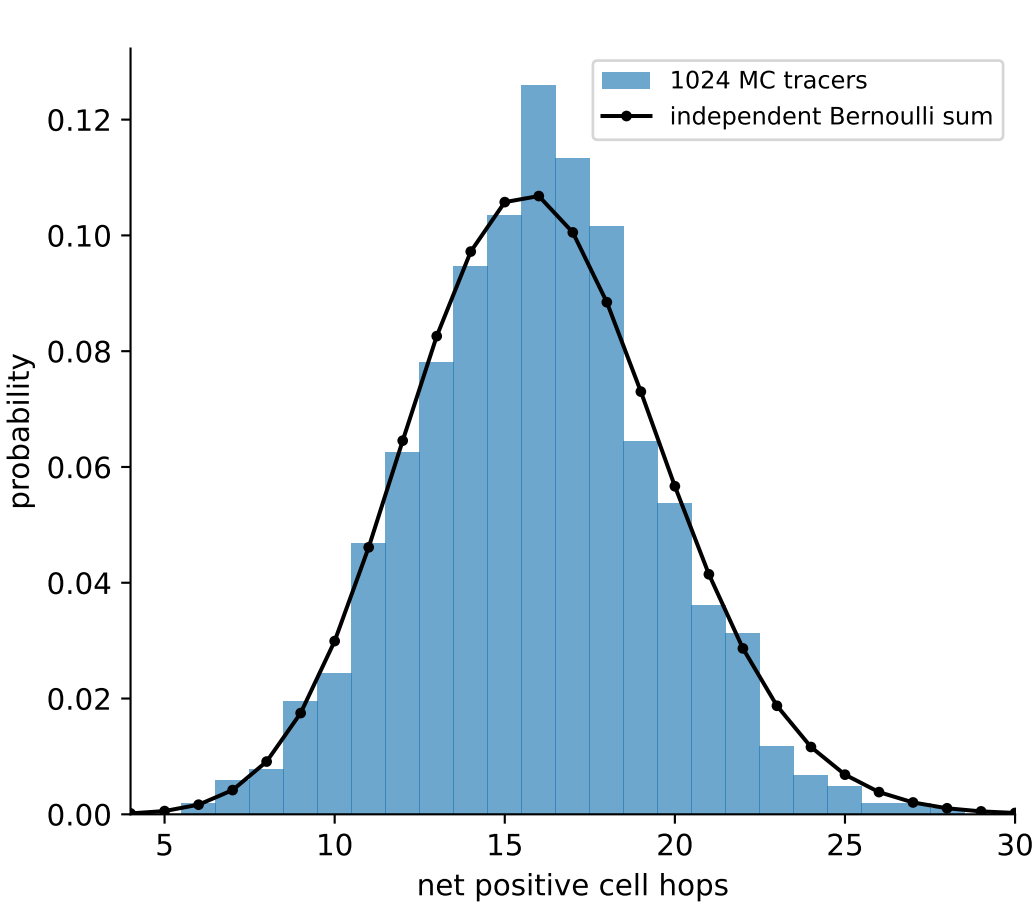

First-inside event strength: bias and spread against independent input Mach
Pressure Mach − input Mach
Temperature Mach − input Mach
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
seed11
seed29
seed47
seed83
seed127
seed211
seed307
seed401
mach1.2
mach1.5
mach3
mach5
gamma53
stationary
boost2
reflected
nx64
nx256
cfl0.15
cfl0.45
planar3d

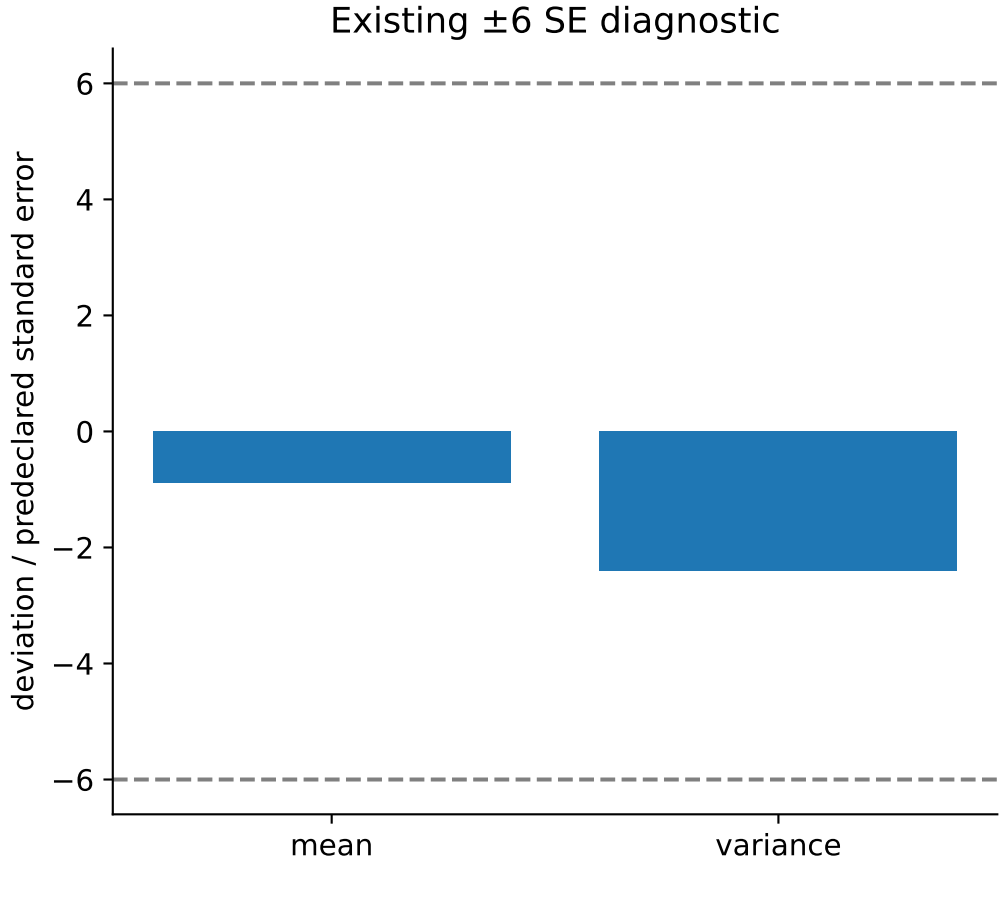

Resolution and cadence change recorded timing brackets
cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
0.0004
0.0006
0.0008
0.0010
0.0012
0.0014
entry bracket width (code time)
nx64
seed11
nx256
cfl0.15
cfl0.45
cumulative fraction
exit bracket width (code time)

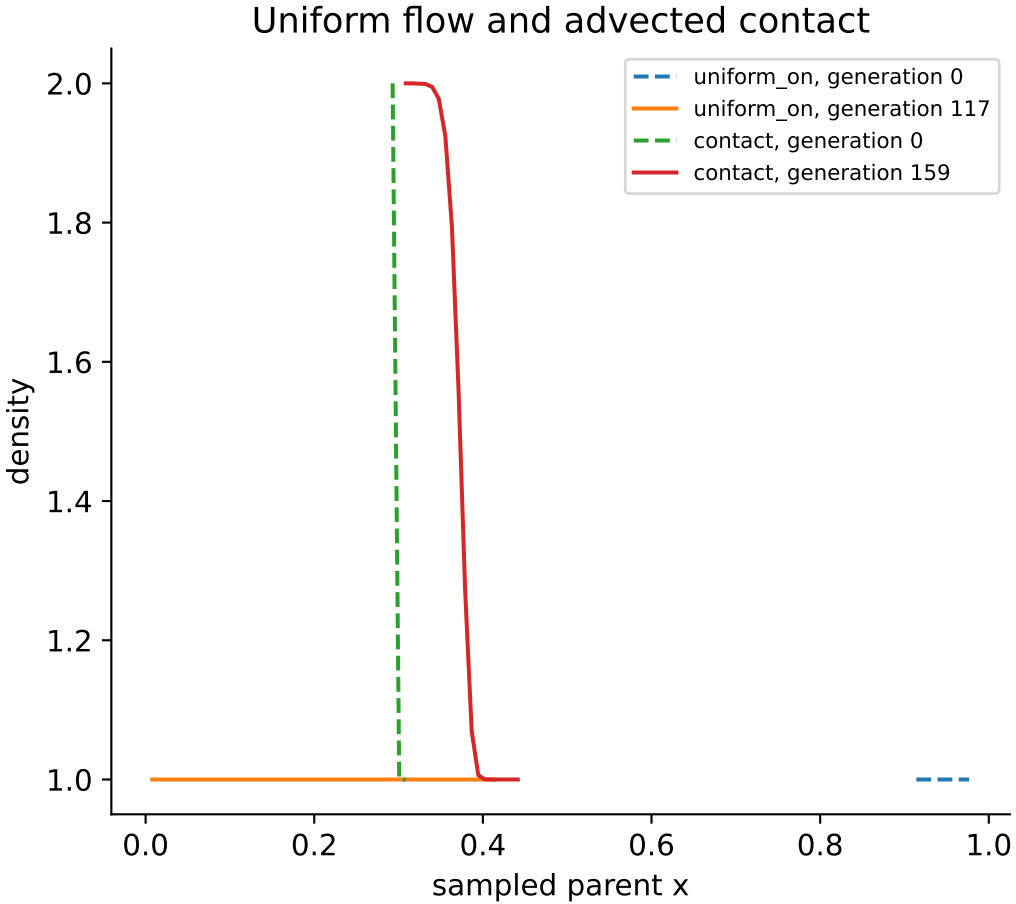

seed11: blue first episode; red rearmed episode; × suspended
particle ID
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
0.08
0.10
0.12
0.14
0.16
0.18
0.20
0.22
0.24
code time

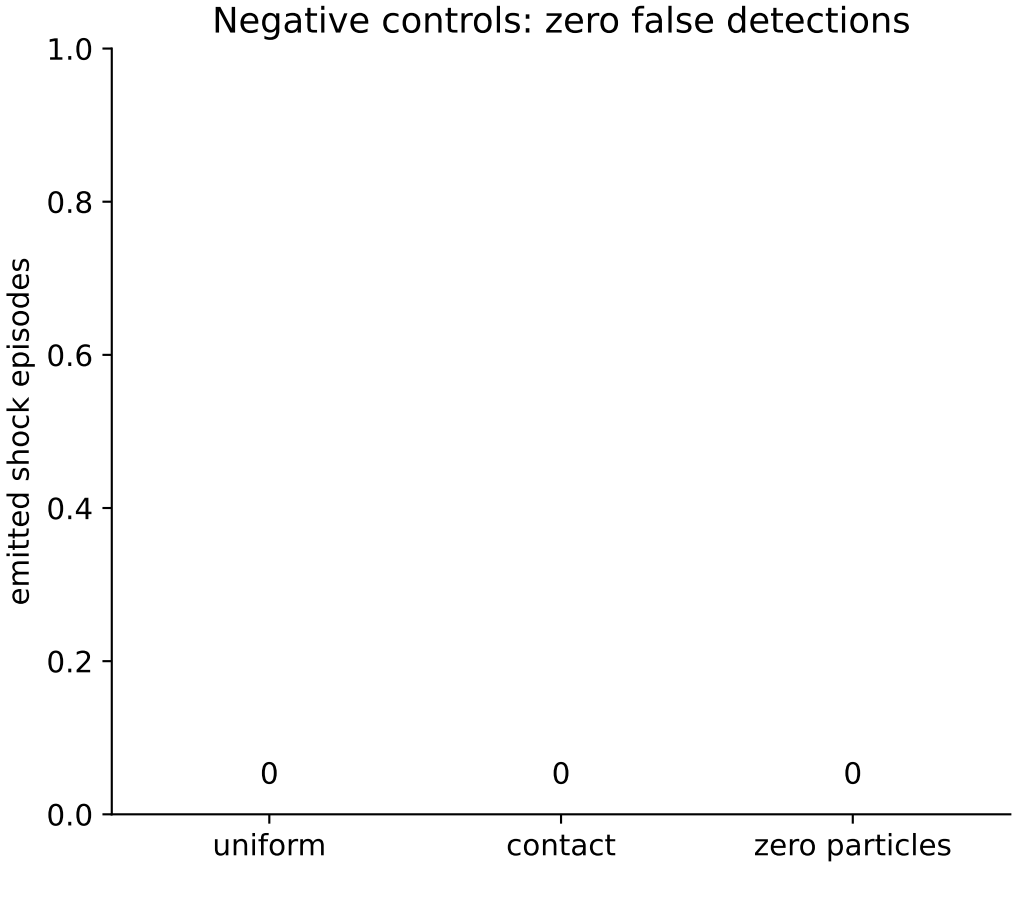

mach5: blue first episode; red rearmed episode; × suspended
particle ID
115
74
57
47
43
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
code time

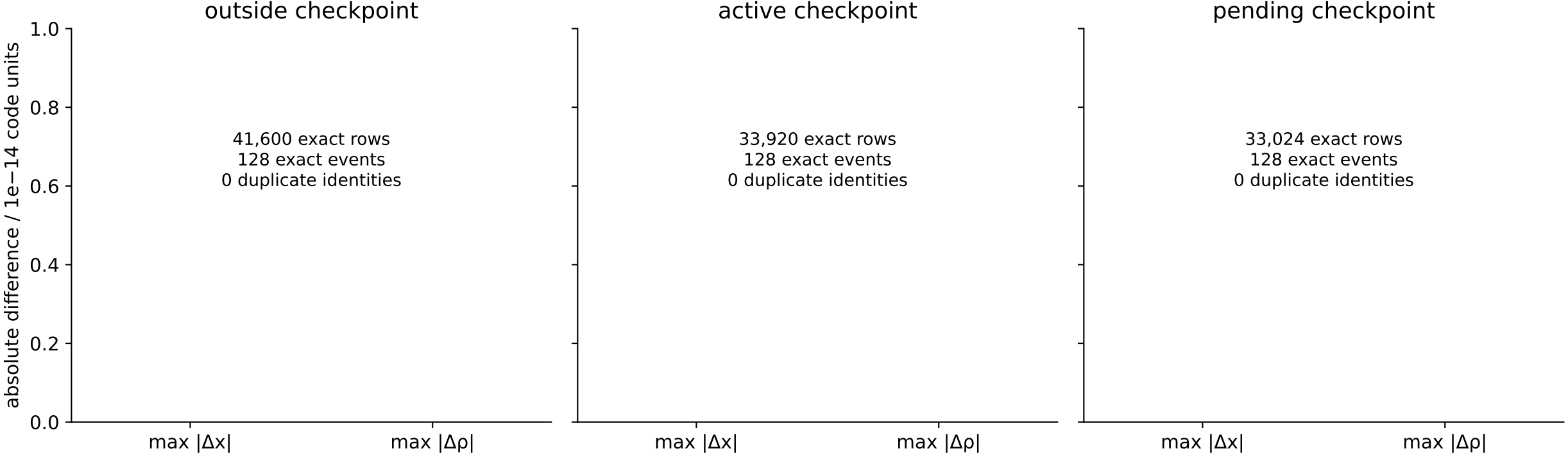

boost2: blue first episode; red rearmed episode; × suspended
particle ID
126
102
80
67
52
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
0
×
0.075
0.100
0.125
0.150
0.175
0.200
0.225
0.250
code time

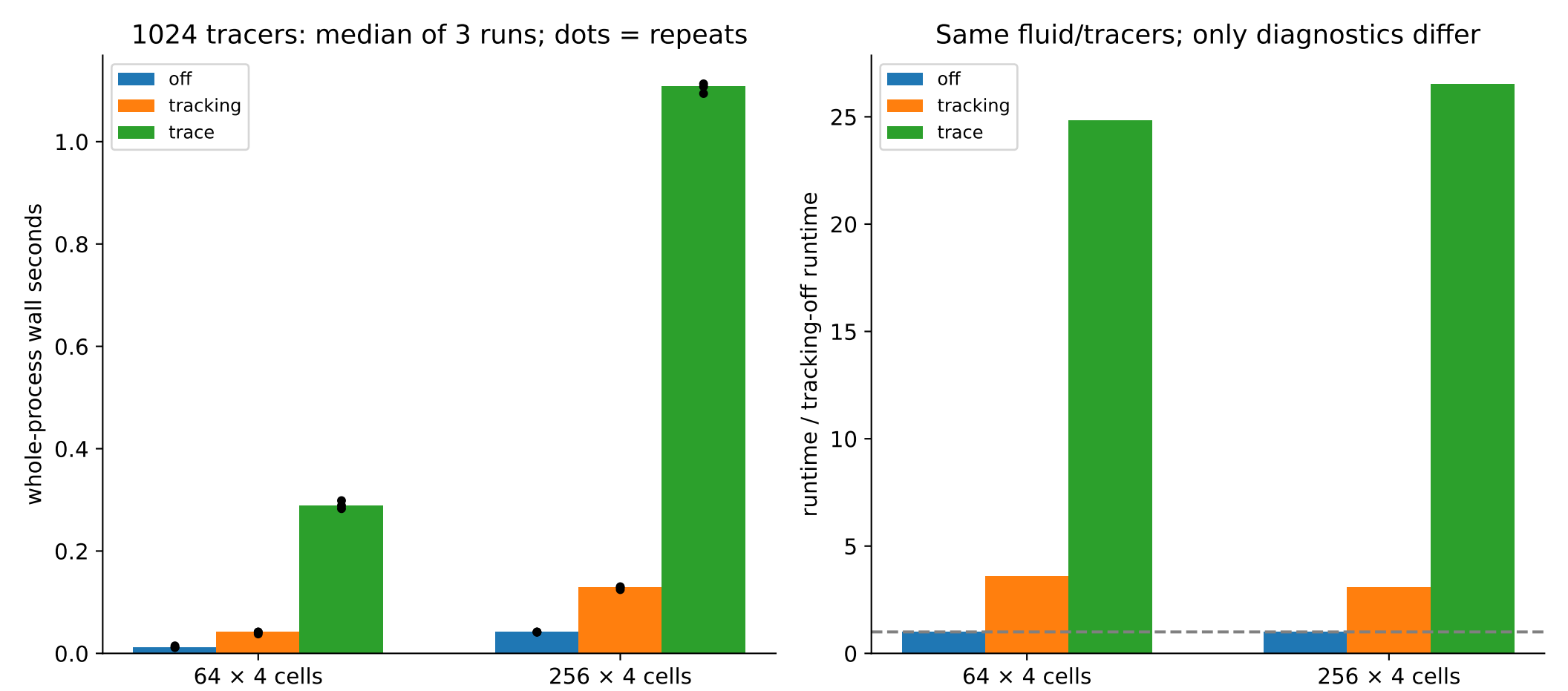

1024 MC tracers
independent Bernoulli sum
probability
0.00
0.02
0.04
0.06
0.08
0.10
0.12
5
10
15
20
25
30
net positive cell hops

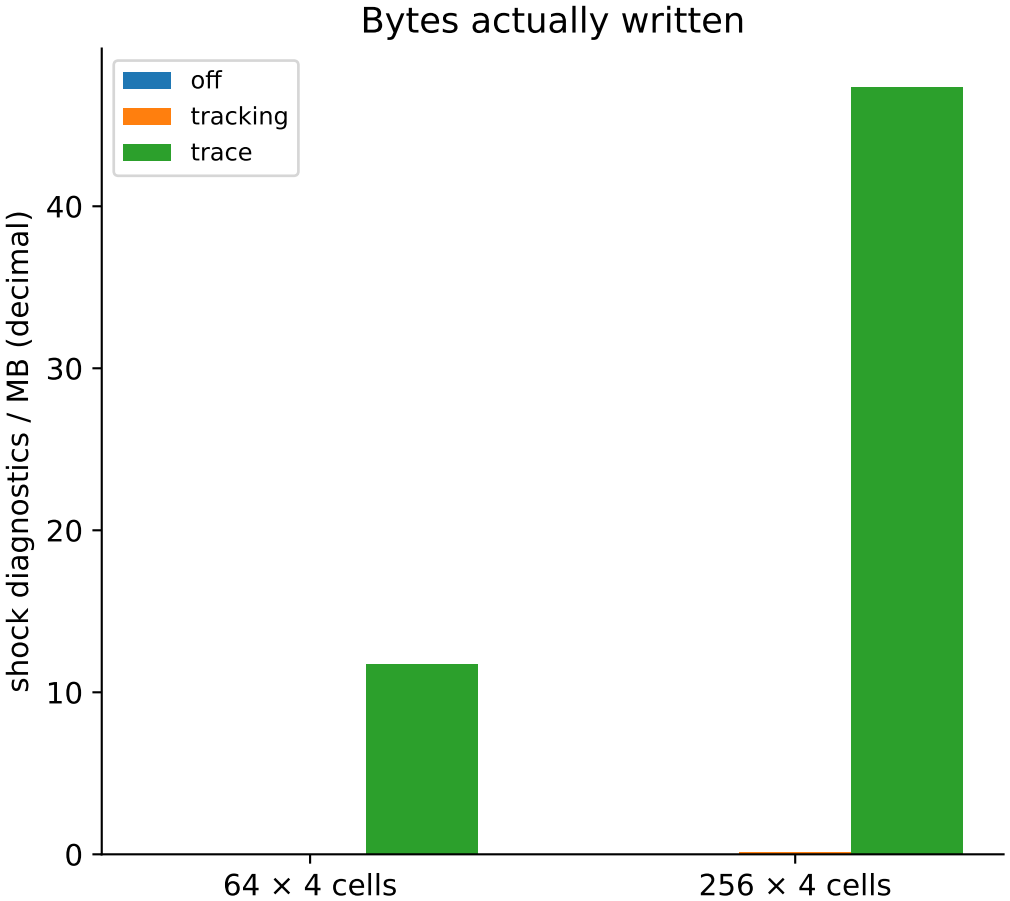

Existing ±6 SE diagnostic
deviation / predeclared standard error
6
4
2
0
−2
−4
−6
mean
variance

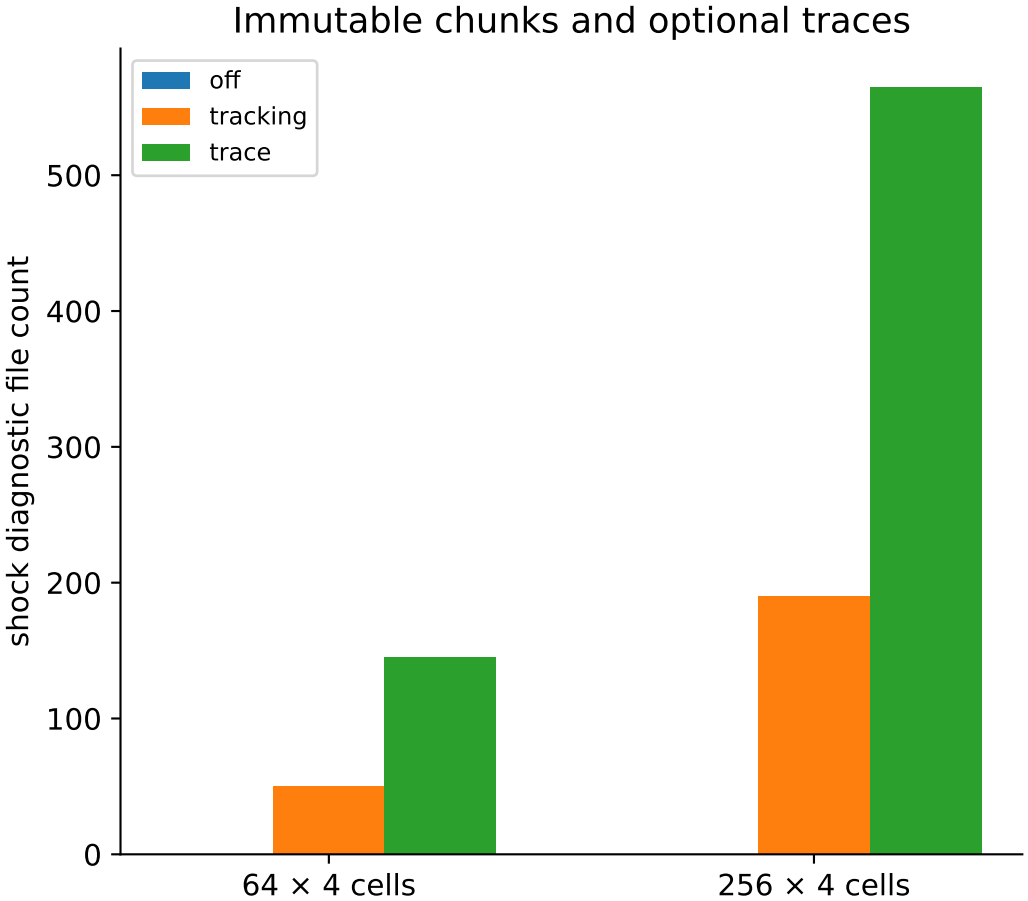

Uniform flow and advected contact
uniform_on, generation 0
uniform_on, generation 117
contact, generation 0
contact, generation 159
density
2.0
1.8
1.6
1.4
1.2
1.0
0.0
0.2
0.4
0.6
0.8
1.0
sampled parent x


Negative controls: zero false detections
emitted shock episodes
1.0
0.8
0.6
0.4
0.2
0.0
0
0
0
uniform
contact
zero particles


outside checkpoint
active checkpoint
pending checkpoint
absolute difference / 1e−14 code units
1.0
0.8
0.6
0.4
0.2
0.0
41,600 exact rows
128 exact events
0 duplicate identities
33,920 exact rows
128 exact events
0 duplicate identities
33,024 exact rows
128 exact events
0 duplicate identities
max |Δx|
max |Δρ|
max |Δx|
max |Δρ|
max |Δx|
max |Δρ|


1024 tracers: median of 3 runs; dots = repeats
whole-process wall seconds
off
tracking
trace
0.0
0.2
0.4
0.6
0.8
1.0
64 × 4 cells
256 × 4 cells
Same fluid/tracers; only diagnostics differ
runtime / tracking-off runtime
off
tracking
trace
0
5
10
15
20
25
64 × 4 cells
256 × 4 cells


Bytes actually written
off
tracking
trace
shock diagnostics / MB (decimal)
0
10
20
30
40
64 × 4 cells
256 × 4 cells


Immutable chunks and optional traces
off
tracking
trace
shock diagnostic file count
0
100
200
300
400
500
64 × 4 cells
256 × 4 cells